DAY 4 · THURSDAY 19 NOVEMBER 2026

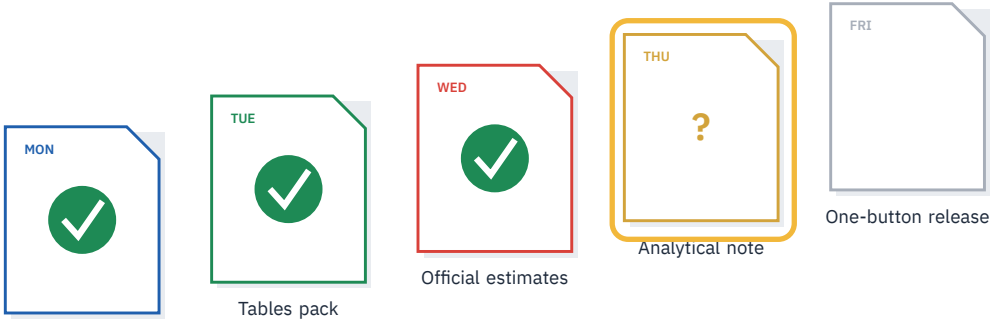
From estimates to explanations

Tests, regression models and price series



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National Bureau of Statistics, Seychelles
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Five days, five products



YOUR TURN · TASK 1

Download today’s files

DO

Type, one line at a time:

```
cd "C:/NBS_Stata"
net from "course address/day4"
net get day4
```

YOU SHOULD SEE

A list of files copied into `day4`: do-files, labs, catch-up files.

IF NOT

No internet? Copy the `day4` folder from the USB stick into `C:\NBS_Stata`.

Yesterday: how many are unemployed. Today: who is more likely to be, and is the difference real?

THURSDAY MORNING AT NBS

The Minister’s office asks three questions

QUESTION 1

Do women in employment work fewer hours than men? Is the gap real or sampling noise?

QUESTION 2

Who is more likely to be unemployed: the young, women, people on Praslin?

QUESTION 3

Is inflation rising again? Which prices are driving it?

By the end of today one do-file answers all three and writes a two-page note in Word.

What you will be able to do today

AFTER THIS PART YOU CAN

- test whether two groups really differ, the way NBS should report it
- fit and read a regression with categorical characteristics
- check a model and choose the right standard errors
- show who is more likely to be unemployed, in percentage points
- turn a monthly price index into inflation, and chart it



1

DAY 4 · PART 1

Is the difference real?

THE PROBLEM

Employed women report fewer hours than men. Could that be luck?

WHAT WE HAVE

2,999 employed persons interviewed, with their usual weekly hours.

WHAT WE NEED

A statement about all employed people in Seychelles, not only those interviewed.

WHY IT IS HARD

Another sample would give other means. A gap must be larger than that noise to count.

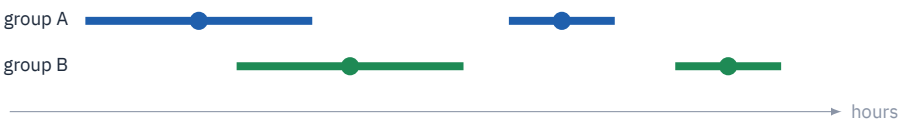
A test and a confidence interval measure the noise.

Part 1: comparing two groups

AFTER THIS PART YOU CAN

- give a mean with its 95% interval (`ci means`)
- compare two means (`ttest`) and two shares (`prtest`)
- say why NBS reports the design-based version
- get a design-based difference with `lincom` and `test`

What a 95% interval says



intervals overlap:
the gap may be noise

far apart:
the gap is real

The dot is the estimate. The bar is the range that contains the true value in 95 of 100 samples drawn the same way.

YOUR TURN · TASK 2

Open yesterday's file and rebuild the flags

DO

Add to `my_day4.do` and run:

```
use "data/lfs_svy.dta", clear
generate wap = (ilo_age >= 15)
generate lf = (ilo_lfs19 == 1 | ilo_lfs19 == 2)
generate unemp = (ilo_lfs19 == 2)
generate emp = (ilo_lfs19 == 1)
svyset
```

YOU SHOULD SEE

Four silent `generate` lines, then the design: weight `wgt_final`, strata `district`, PSU `psu`.

IF NOT

`variable lf already defined`? You ran it twice: start again from `use`.

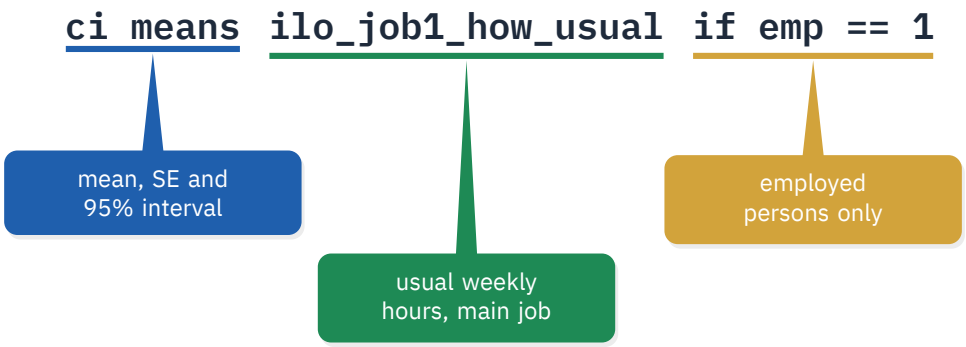
The design came with the file

```
. svyset

Sampling weights: wgt_final (1)
             VCE: linearized
      Subpop. df: off
     Single unit: missing
        Strata 1: district (2)
 Sampling unit 1: psu (3)
           FPC 1: <zero>
```

1. the weight NBS gave every person
2. districts as strata
3. PSUs as clusters

A mean with its interval:



YOUR TURN · TASK 3

Usual hours of the employed

DO

Type:

```
ci means ilo_job1_how_usual if emp == 1
```

YOU SHOULD SEE

One row: number of employed persons, mean hours, SE and the interval.

IF NOT

`variable emp not found`? Run Task 2 again.

EXTRA Is 40 hours inside the interval?

Read the interval

```
. ci means ilo_job1_how_usual if emp == 1

    Variable |        Obs        Mean    Std. err.       [95% conf. interval]
-------------+---------------------------------------------------------------
~1_how_usual |      2,999    39.82861    .2096843        39.41747    40.23975
```

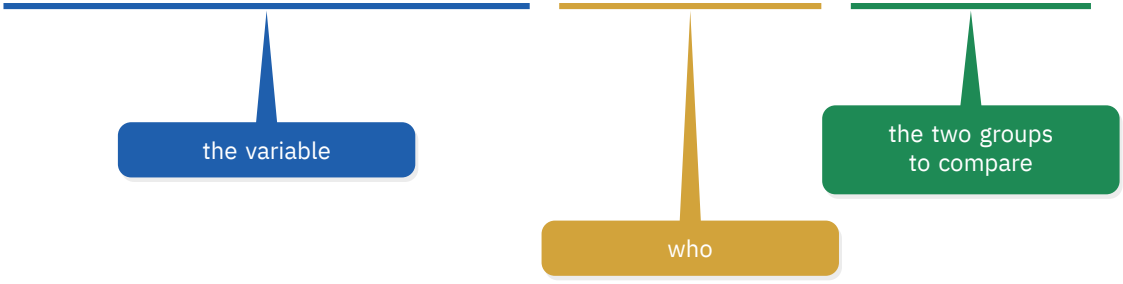
1

1 2,999 employed persons; 39.8 hours on average, between 39.4 and 40.2

An interval is a statement about all employed people, not about the 2,999 we met.

Compare two means:

```
ttest ilo_job1_how_usual if emp == 1, by(ilo_sex)
```



YOUR TURN · TASK 4

Hours of women and men

DO

Type:

```
ttest ilo_job1_how_usual if emp == 1, by(ilo_sex)
```

YOU SHOULD SEE

Two rows (men, women), a row for both, the difference, and three p-values at the bottom.

IF NOT

Red text about groups? Check you typed `by(ilo_sex)`.

EXTRA Which group has the larger standard deviation?

Read the test

```
. ttest ilo_job1_how_usual if emp == 1, by(ilo_sex)
------------------------------------------------------------------------------
   Group |     Obs        Mean    Std. err.   Std. dev.   [95% conf. interval]
---------+--------------------------------------------------------------------
1 - Male |   1,431     41.3522    .3436461    12.99964     40.6781    42.02631  (1)
2 - Fema |   1,568    38.43814    .2448481    9.695491    37.95787     38.9184  (2)
---------+--------------------------------------------------------------------
Combined |   2,999    39.82861    .2096843    11.48297    39.41747    40.23975
---------+--------------------------------------------------------------------
    diff |            2.914064    .4164891                2.09743    3.730697  (3)
------------------------------------------------------------------------------
    diff = mean(1 - Male) - mean(2 - Fema)                        t =   6.9967
H0: diff = 0                                     Degrees of freedom =     2997

    Ha: diff < 0                 Ha: diff != 0                 Ha: diff > 0
 Pr(T < t) = 1.0000         Pr(|T| > |t|) = 0.0000          Pr(T > t) = 0.0000  (4)
```

1. men: 41.4 hours on average
2. women: 38.4
3. gap 2.9 hours, interval 2.1 to 3.7
4. middle p-value 0.0000: not chance

p below 0.05: the gap is larger than sampling noise.

Here p is 0.0000: women in employment report fewer usual hours than men.

The interval says how large: 2.1 to 3.7 hours a week.

COMMON ERROR

Three groups in a two-group test

```
. ttest ilo_job1_how_usual if emp == 1, by(region)
more than 2 groups found, only 2 allowed
```

WHY

`ttest` compares exactly two groups. `region` has seven.

FIX

Pick two with `if`: `if inlist(region, 6, 7)`. Many groups at once: regression, Part 2.

YOUR TURN · TASK 5

A variation: unequal spreads

DO

Type:

```
ttest ilo_job1_how_usual if emp == 1, by(ilo_sex)
unequal
```

YOU SHOULD SEE

The same means; a slightly different SE of the difference and degrees of freedom with decimals.

IF NOT

EXTRA Which SE is larger, with or without `unequal`?

Equal or unequal spreads

EQUAL (DEFAULT)

```
diff |   2.914064   .4164891                2.09743   3.730697
```

1

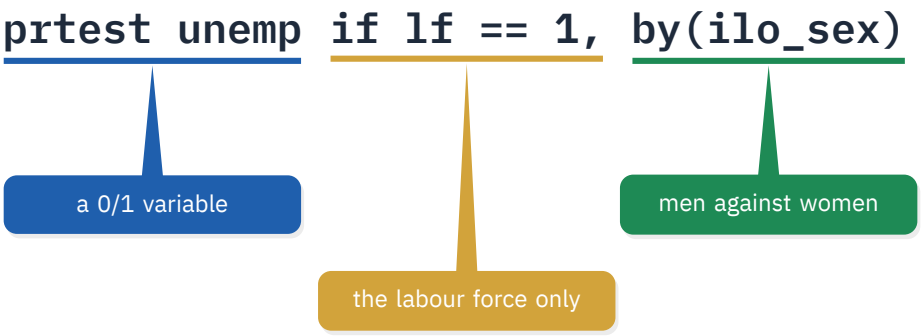
UNEQUAL

```
diff |   2.914064   .4219517                2.086673   3.741454
```

2

1 SE of the gap .4165 2 .4220. Same conclusion: the gap is 2.9 hours, clearly above noise.

Compare two shares:



YOUR TURN · TASK 6

Unemployment of women and men

DO

Type:

```
prtest unemp if lf == 1, by(ilo_sex)
```

YOU SHOULD SEE

Two shares, their difference and its interval, three p-values.

IF NOT

`unemp` must be 0/1: rebuild it as in Task 2.

EXTRA Is the gap above or below one percentage point?

Read the test of shares

```
. prtest unemp if lf == 1, by(ilo_sex)
------------------------------------------------------------------------------
       Group |       Mean   Std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
    1 - Male |   .0367647   .0048654                      .0272288    .0463007  (1)
  2 - Female |   .0361963   .0046263                       .027129    .0452637  (2)
-------------+----------------------------------------------------------------
        diff |   .0005684   .0067137                     -.0125903    .0137271  (3)
             |  under H0:   .0067116     0.08   0.933
   Ha: diff < 0                 Ha: diff != 0                 Ha: diff > 0
 Pr(Z < z) = 0.5337         Pr(|Z| > |z|) = 0.9325          Pr(Z > z) = 0.4663  (4)
```

1. men: 3.68% of the labour force interviewed
2. women: 3.62%
3. gap .06 points, interval from −1.3 to 1.4
4. p = 0.93: no evidence of a gap

WARNING

`ttest` and `prtest` treat the file as a simple random sample.

No weights, no strata, no PSUs. NBS does not publish them.

For NBS: `svy:` estimate, then `lincom`.

The design-based difference:

```
svy, subpop(lf): mean unemp, over(ilo_sex)  lincom
```

yesterday's rates by sex

a combination of the estimates, with its SE

Name the two rates

```
lincom c.unemp@1.ilo_sex - c.unemp@2.ilo_sex
```

- `c.unemp@1.ilo_sex`: the rate of `unemp` for sex = 1 (men)
- `c.unemp@2.ilo_sex`: the same for women
- the minus sign: men minus women

YOUR TURN · TASK 7

The design-based gap

DO

Type:

```
svy, subpop(lf): mean unemp, over(ilo_sex)
lincom c.unemp@1.ilo_sex - c.unemp@2.ilo_sex
```

YOU SHOULD SEE

Yesterday's two rates, then one row: the difference, its SE, t, p and interval.

IF NOT

`not found`? Copy the name from `mean`, `coeflegend` (Task 9).

Read the design-based gap

```
. svy, subpop(lf): mean unemp, over(ilo_sex)
c.unemp@ilo_sex |
      1 - Male  |      .03455   .0061649      .0223439    .0467561  (1)
    2 - Female  |   .0317034   .0045353      .0227238     .040683  (2)
. lincom c.unemp@1.ilo_sex - c.unemp@2.ilo_sex
------------------------------------------------------------------------------
        Mean | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
         (1) |   .0028466   .0082104     0.35   0.729    -.0134094    .0191025  (3)
------------------------------------------------------------------------------
```

1. men 3.46%
2. women 3.17%: weighted, design-based

3. gap .28 points, interval from −1.3 to 1.9; p = 0.73

YOUR TURN · TASK 8

Where the names come from

DO

Type:

```
mean, coeflegend
```

YOU SHOULD SEE

The same table, with a Legend column: the exact name of each estimate.

IF NOT

The legend

```
. mean, coeflegend
------------------------------------------------------
                |       Mean   Legend
----------------+-------------------------------------
c.unemp@ilo_sex |
      1 - Male  |     .03455  _b[c.unemp@1bn.ilo_sex]  1
    2 - Female  |   .0317034  _b[c.unemp@2.ilo_sex]  2
------------------------------------------------------
```

1. `1bn`: the base level, ”bn” = base none
2. copy these names into `lincom` or `_b[ ]`

COMMON ERROR

A level that does not exist

```
. lincom c.unemp@1.ilo_sex - c.unemp@3.ilo_sex
regressor I3.ilo_sex#C.unemp not found
```

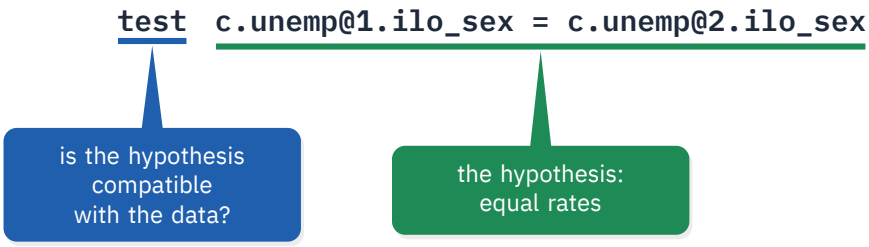
WHY

`ilo_sex` has two levels, 1 and 2. There is no rate for 3.

FIX

Run `mean, coeflegend` and copy the names.

The same question as a p-value:



YOUR TURN · TASK 9

Test equal rates

DO

Type:

```
test c.unemp@1.ilo_sex = c.unemp@2.ilo_sex
```

YOU SHOULD SEE

An F statistic and Prob > F.

IF NOT

Read the test

```
. test c.unemp@1.ilo_sex = c.unemp@2.ilo_sex

Adjusted Wald test

 ( 1)  c.unemp@1bn.ilo_sex - c.unemp@2.ilo_sex = 0

       F(  1,    120) =    0.12 (1)
            Prob > F =    0.7294 (2)
```

1. F with 1 and 120 degrees of freedom: 120 is the design df
2. p = 0.73, the same p as `lincom`

Two answers to one question

PRTEST: INTERVIEWS

```
diff |   .0005684   .0067137                -.0125903    .0137271
```

1

DESIGN: SEYCHELLES

```
(1) |   .0028466   .0082104     0.35  0.729    -.0134094    .0191025
```

2

1 gap .06 points, SE .0067 2 gap .28 points, SE .0082. Different estimate, wider interval.
Both: no evidence of a gap.

For NBS figures: `svy:` estimate, then `lincom`.

ON YOUR OWN · A

Women’s and men’s hours, the NBS way

THE QUESTION

Estimate the design-based mean usual hours of employed men and women, and their difference with its interval.

HINT

`svy, subpop(emp): mean ... , over(ilo_sex)`, then `lincom`.

ON YOUR OWN · A · ANSWER

```
svy, subpop(emp): mean ilo_job1_how_usual,
over(ilo_sex)
lincom c.ilo_job1_how_usual@1.ilo_sex - ///
c.ilo_job1_how_usual@2.ilo_sex
```

```
c.ilo_job1_how_usual@ilo_sex |
                  1 - Male  |    41.2178   .5167441      40.19468    42.24092
                2 - Female  |   38.16913    .404831      37.36759    38.97066
-------------------------------------------------------------------------------
        Mean | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+-----------------------------------------------------------------
         (1) |   3.048675    .460111     6.63   0.000     2.137687    3.959663
-------------------------------------------------------------------------------
```

1. men 41.2, women 38.2 hours
2. gap 3.0 hours, 2.1 to 4.0

ON YOUR OWN · B

Young against 25-34

THE QUESTION

How many points higher is unemployment among the 15-24 than among the 25-34? Give the interval.

HINT

`over(agegrp)`; the levels are 1 to 4.

ON YOUR OWN · B · ANSWER

```
svy, subpop(lf): mean unemp, over(agegrp)
lincom c.unemp@1.agegrp - c.unemp@2.agegrp
```

```
c.unemp@agegrp |
        15-24  |   .1277627   .0212881      .0856137    .1699118
        25-34  |   .0300666   .0068401      .0165236    .0436095
        35-54  |   .0194959   .0040849      .0114081    .0275837
          55+  |   .0131063   .0048821      .0034402    .0227724
        Mean | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+------------------------------------------------------------------
         (1) |   .0976962   .0227299     4.30   0.000     .0526926    .1426997
```

1. 15-24: 12.8%; 25-34: 3.0%
2. 9.8 points higher, 5.3 to 14.3

QUICK CHECK

prtest gives p = 0.93; the design-based lincom gives p = 0.73. Which do you report for NBS?

A. prtest: it is the test for shares

B. lincom after svy: it uses weights, strata and PSUs

C. neither: they disagree

ANSWER

prtest gives p = 0.93; the design-based lincom gives p = 0.73. Which do you report for NBS?

A prtest: it is the test for shares

B lincom after svy: it uses weights, strata and PSUs

C neither: they disagree

Both say ”no evidence of a gap”. Only the design-based one describes Seychelles.

Part 1 on one page

CHEAT CARD · KEEP THIS ONE

`ci means y if emp == 1`	mean with 95% interval (no design)
`ttest y, by(g)`	two means, quick look
`prtest flag, by(g)`	two shares, quick look
`svy, subpop(s): mean y, over(g)`	design-based means by group
`mean, coeflegend`	the names of the estimates
`lincom c.y@1.g - c.y@2.g`	the gap with SE and interval
`test c.y@1.g = c.y@2.g`	the same as a p-value

Practice on your own

PRACTICE SET · practice_D4.do, PART 1

CORE	P1	`ci means` of age for the unemployed. Type the upper limit.
CORE	P2	`ttest` of usual hours, public against private sector. Type the gap.
CORE	P3	Design-based participation by sex; the gap with `lincom`.
EXTENSION	P4	Design-based unemployment, Praslin against North (`lincom`).
CHALLENGE	P5	One `test` line: are the four age-group rates all equal?

Each exercise has a CHECK line: silent means correct. Solutions are released at the end of the day.

You can say whether a difference between two groups is real, the way NBS should report it.

lost? open `catchup_D4_S1.do`



2

DAY 4 · PART 2

Many differences at once

THE PROBLEM

Women work fewer hours. But are they also younger, in other regions, with other schooling?

WHAT WE HAVE

One gap: 2.9 hours, men against women.

WHAT WE NEED

The gap between women and men of the same age, region and education.

WHY TESTS CANNOT

A test compares two groups. Here four characteristics act together.

Regression holds the other characteristics fixed.

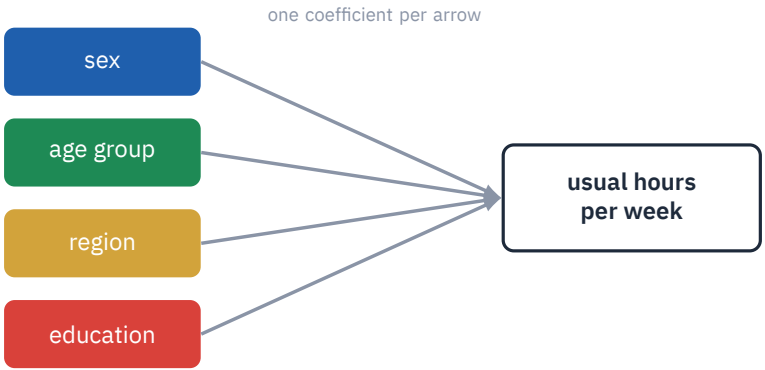
Part 2: linear regression

AFTER THIS PART YOU CAN

- fit `regress` and read coefficient, interval and R-squared
- use categorical characteristics with `i.` and choose the base with `ib#.`
- add a curve with `c.age##c.age`
- save fitted values and residuals with `predict`

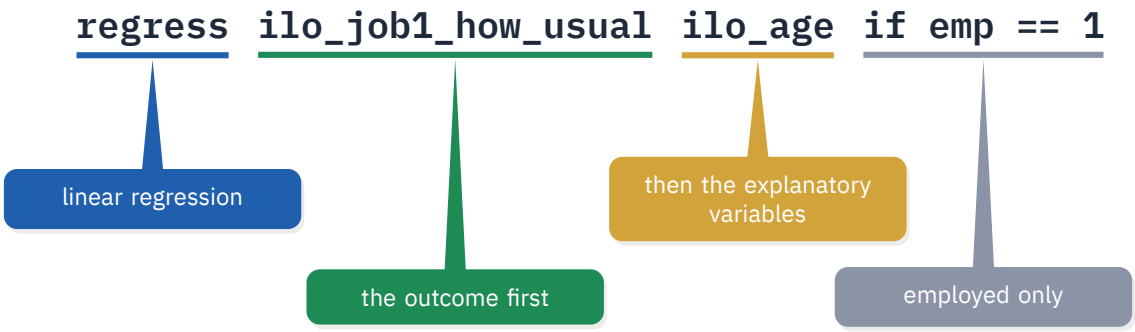
A regression compares groups while holding the other characteristics fixed.

Hours explained by four characteristics



Each coefficient: the change in hours for that characteristic, the others held fixed.

One explanatory variable:



YOUR TURN · TASK 10

Hours and age

DO

Type:

```
regress ilo_job1_how_usual ilo_age if emp == 1
```

YOU SHOULD SEE

An analysis-of-variance table at the top, then two rows: `ilo_age` and `_cons`.

IF NOT

EXTRA Is the age coefficient positive or negative?

Read a regression

```
. regress ilo_job1_how_usual ilo_age if emp == 1

      Source |       SS           df       MS      Number of obs   =     2,999
-------------+----------------------------------   F(1, 2997)      =      0.60
       Model |  78.6794781         1  78.6794781   Prob > F        =    0.4399
    Residual |  395233.226     2,997  131.876285   R-squared       =    0.0002
-------------+----------------------------------   Adj R-squared   =   -0.0001
       Total |  395311.905     2,998  131.858541   Root MSE        =    11.484

------------------------------------------------------------------------------
~1_how_usual | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
     ilo_age |  -.0124449   .0161118    -0.77   0.440    -.0440362    .0191464
       _cons |   40.35037   .7072986    57.05   0.000     38.96353    41.73721
------------------------------------------------------------------------------
```

1. 2,999 employed persons
2. R-squared .0002: age explains almost nothing of hours
3. one year older: .01 hours fewer, p = 0.44
4. the line at age 0: a starting point only

A coefficient is a slope.

Here: one year older, .0124 hours fewer a week, with an interval from −.044 to .019 that includes zero.

Age alone does not explain hours.

Categories in a regression:



What `i.region` does

	2.region	3.region	4.region	5.region	6.region	7.region	
1 North	0	0	0	0	0	0	base: all zeros
2 Central 1	1	0	0	0	0	0	
3 Central 2	0	1	0	0	0	0	
4 EastSouth	0	0	1	0	0	0	
5 West	0	0	0	1	0	0	
6 Praslin & LD	0	0	0	0	1	0	
7 Perseverance	0	0	0	0	0	1	

Stata makes six 0/1 variables in memory. Each coefficient compares one region with the base, North.

YOUR TURN · TASK 11

Sex, age group and region together

DO

Type:

```
regress ilo_job1_how_usual i.ilo_sex i.agegrp
i.region if emp == 1
```

YOU SHOULD SEE

One row for women, three for age groups, six for regions, and the constant.

IF NOT

`factor-variable ... may not contain noninteger values`? Check you typed `i.` only before categories.

EXTRA Which region has the lowest coefficient?

Read the categories

```
. regress ilo_job1_how_usual i.ilo_sex i.agegrp i.region if emp == 1
    Residual |  386355.613     2,988  129.302414   R-squared       =    0.0227
-------------------------------------------------------------------------------
   ilo_job1_how_usual | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
----------------------+--------------------------------------------------------------
             ilo_sex |
         2 - Female  |  -2.941233   .4168892    -7.06   0.000    -3.758652   -2.123814  1
              region |
       2- Central 1  |   1.126683   .7321685     1.54   0.124    -.3089227    2.562288  2
6- Praslin & LaDigue  |  -1.916317   .8517556    -2.25   0.025    -3.586404   -.2462303  3
     7- Perseverance  |   1.755887    1.00871     1.74   0.082    -.2219494    3.733724
                      |
               _cons  |   41.20147   .8329428    49.46   0.000     39.56827    42.83467  4
```

1. women: 2.94 fewer, same age group and region
2. Central 1 vs North: +1.1, includes 0
3. Praslin & La Digue: 1.9 fewer, p = 0.025
4. men, 15-24, North: 41.2 hours

Women work 2.9 hours fewer than men of the same age group and region.

The gap barely moved from the raw 2.9: age and region do not explain it.

That is the finding the Minister asked for.

Choose the base:

`ib7.region`

b7: category 7 (Perseverance) is the base

YOUR TURN · TASK 12

Perseverance as the base

DO

Type:

```
regress ilo_job1_how_usual i.ilo_sex i.agegrp
ib7.region if emp == 1
```

YOU SHOULD SEE

The same model; now North appears and Perseverance is missing.

IF NOT

EXTRA Did R-squared change?

Same model, another base

```
. regress ilo_job1_how_usual i.ilo_sex i.agegrp ib7.region if emp == 1
    Residual |  386355.613     2,988  129.302414   R-squared       =    0.0227 (1)
             region |
           1- North  |  -1.755887    1.00871    -1.74   0.082    -3.733724    .2219494 (2)
       2- Central 1  |  -.6292045    1.02215    -0.62   0.538    -2.633394    1.374985
       3- Central 2  |  -2.426333   1.048447    -2.31   0.021    -4.482084   -.3705824
       4- EastSouth  |  -1.637214   .9670974    -1.69   0.091    -3.533458    .2590303
         5- West     |  -2.183337   1.019684    -2.14   0.032    -4.182691    -.183983
6- Praslin & LaDigue |  -3.672204   1.110495    -3.31   0.001    -5.849617   -1.494792 (3)
                     |
               _cons |   42.95736   1.088398    39.47   0.000     40.82327    45.09144
```

1. R-squared .0227: the fit is the same
2. North against Perseverance: 1.8 fewer
3. Praslin against Perseverance: 3.7 fewer, p = 0.001

COMMON ERROR

A text variable in a regression

```
. regress ilo_job1_how_usual unique_ind if emp == 1
no observations
```

WHY

`unique_ind` is a string. `regress` drops every row with a non-number, so none are left.

FIX

Regress on numeric variables. Identifiers are never explanatory variables.

It runs, but it is wrong: `i.ilo_age`

```
. regress ilo_job1_how_usual i.ilo_age if emp == 1
      Source |       SS           df       MS      Number of obs   =     2,999
-------------+----------------------------------   F(65, 2933)     =      1.02  (1)
       Model |  8705.92235         65  133.937267   Prob > F        =    0.4416
    Residual |  386605.983     2,933  131.812473   R-squared       =    0.0220
-------------+----------------------------------   Adj R-squared   =    0.0003  (2)
       Total |  395311.905     2,998  131.858541   Root MSE        =    11.481
------------------------------------------------------------------------------
~1_how_usual | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
     ilo_age |
          16 |       -5.5   12.40086    -0.44   0.657    -29.81526    18.81526  (3)
          17 |  -1.111111     12.102    -0.09   0.927    -24.84038    22.61816
       _cons |         40   11.48096     3.48   0.001     17.48844    62.51156
```

(1) 65 coefficients, one per age (2) adjusted R-squared .0003 (3) SE about 12 hours each. Use `agegrp`, or age as a number (next).

YOUR TURN · TASK 13

Look at education first

DO

Type:

```
codebook ilo_edu_aggregate, tabulate(10)
```

YOU SHOULD SEE

Five levels from ”Less than basic” to ”Level not stated”.

IF NOT

EXTRA Which level is the base if you type `i.`?

Five levels, one tiny

Tabulation: Freq.	Numeric	Label
146	1	1 - Less than basic
1,651	2	2 - Basic
3,275	3	3 - Intermediate
329	4	4 - Advanced
487	5	5 - Level not stated

1. 146 persons, and how many employed? Part 3 finds out
2. the largest group: 3,275

YOUR TURN · TASK 14

Add education

DO

Type:

```
regress ilo_job1_how_usual i.ilo_sex i.agegrp
i.region ///
    i.ilo_edu_aggregate if emp == 1
```

YOU SHOULD SEE

Four education rows, all near −8, with standard errors near 6.6.

IF NOT

Read education

```
. regress ilo_job1_how_usual i.ilo_sex i.agegrp i.region i.ilo_edu_aggregate if emp == 1
          2 - Female  |  -2.927601   .4179575    -7.00   0.000    -3.747115   -2.108087
           2 - Basic  |  -7.716295    6.61439    -1.17   0.243    -20.68552    5.252932
    3 - Intermediate  |  -8.195958   6.579073    -1.25   0.213    -21.09594     4.70402
        4 - Advanced  |  -8.919818   6.609791    -1.35   0.177    -21.88003    4.040392
5 - Level not stated  |  -7.865024   6.701309    -1.17   0.241    -21.00468    5.274629
                      |
                _cons |   49.42621    6.60879     7.48   0.000     36.46796    62.38445
```

1. women: 2.93 fewer, still
2. every level about 8 hours below the base, but SE 6.6: none significant
3. the constant jumps to 49.4: the base group is unusual

Age as a curve:

```
c.ilo_age##c.ilo_age
```

age and age squared, as numbers

YOUR TURN · TASK 15

Hours rise, then fall, with age?

DO

Type:

```
regress ilo_job1_how_usual i.ilo_sex
c.ilo_age##c.ilo_age ///
    i.region i.ilo_edu_aggregate if emp == 1
```

YOU SHOULD SEE

A row for `ilo_age` and one for `c.ilo_age#c.ilo_age`.

IF NOT

EXTRA Is the squared term negative?

Read the curve

```
. regress ilo_job1_how_usual i.ilo_sex c.ilo_age##c.ilo_age i.region i.ilo_edu_aggregate if emp == 1
            ilo_sex |
          2 - Female  |  -2.967413   .4178583    -7.10   0.000    -3.786733   -2.148094
                      |
              ilo_age |   .1877888     .10043     1.87   0.062    -.0091304    .3847079   (1)
                      |
c.ilo_age#c.ilo_age   |  -.0023906    .001183    -2.02   0.043    -.0047102   -.0000709   (2)
```

(1) the slope at age 0: +.19 hours per year

(2) negative, p = 0.043: the curve bends down at older ages

Save what the model says:

```
predict yhat
```

fitted hours
for every row

```
predict res, residuals
```

actual minus fitted

YOUR TURN · TASK 16

Fitted values and residuals

DO

Type:

```
regress ilo_job1_how_usual i.ilo_sex i.agegrp
i.region ///
    i.ilo_edu_aggregate if emp == 1
predict yhat if e(sample)
predict res if e(sample), residuals
summarize ilo_job1_how_usual yhat res if e(sample)
```

YOU SHOULD SEE

Three rows: hours, fitted hours, residuals. The first two have the same mean.

IF NOT

Read the fitted values

```
. summarize ilo_job1_how_usual yhat res if e(sample)

    Variable |        Obs        Mean    Std. dev.       Min        Max
-------------+---------------------------------------------------------
~1_how_usual |      2,999    39.82861    11.48297          1        120
        yhat |      2,999    39.82861    1.766225   35.38136   50.04914
         res |      2,999    1.72e-08    11.34632  -41.53408   78.53034
```

1. actual hours: 1 to 120
2. fitted: 35.4 to 50.0, same mean 39.83
3. residuals average zero

Person by person

	uniqu~nd	ilo_jo..	yhat	res	
1.	1132481	48	39.18854	8.811463	
2.	1132371	35	39.6682	-4.668199	
5.	1132401	45	39.18854	5.811464	
6.	1132451	35	42.35639	-7.356391	
7.	1121181	84	43.01375	40.98625	1

1. reports 84 hours; the model expects 43: residual +41

COMMON ERROR

predict with no model

```
. predict yhat2
last estimates not found
```

WHY

`predict` uses the last estimates. After `ereturn clear` there are none.

FIX

Run the model first. In a do-file, keep `predict` directly after its model.

YOUR TURN · TASK 17

What the model stored

DO

Type:

```
display e(N)
display e(r2)
```

YOU SHOULD SEE

The number of observations and R-squared.

IF NOT

EXTRA `ereturn list` shows all of them.

Stored results

```
. display e(N)
2999 (1)

. display e(r2)
.02365832 (2)
```

1. e(N): 2,999
2. e(r2): .0237, ready to write into a note

ON YOUR OWN · C

Which region works the fewest hours?

THE QUESTION

With Perseverance as the base, which region's employed work the fewest hours, all else equal? By how much?

HINT

`ib7.region`.

ON YOUR OWN · C · ANSWER

```
regress ilo_job1_how_usual i.ilo_sex i.agegrp ///
ib7.region if emp == 1
```

```
              region |
          1- North   |  -1.755887    1.00871    -1.74   0.082    -3.733724    .2219494
       2- Central 1  |  -.6292045    1.02215    -0.62   0.538    -2.633394    1.374985
       3- Central 2  |  -2.426333   1.048447    -2.31   0.021    -4.482084   -.3705824
       4- EastSouth  |  -1.637214   .9670974    -1.69   0.091    -3.533458    .2590303
            5- West  |  -2.183337   1.019684    -2.14   0.032    -4.182691   -.183983
-----------------------+------------------------------------------------------------------
6- Praslin & LaDigue  |  -3.672204   1.110495    -3.31   0.001    -5.849617   -1.494792
```

1

1 Praslin & La Digue, 3.7 hours fewer (1.5 to 5.8)

ON YOUR OWN · D

Does the age curve bend?

THE QUESTION

Replace age group by age and age squared. Is the squared term negative and significant at 5%?

HINT

`c.ilo_age##c.ilo_age`.

ON YOUR OWN · D · ANSWER

```
regress ilo_job1_how_usual i.ilo_sex ///
c.ilo_age##c.ilo_age i.region ///
i.ilo_edu_aggregate if emp == 1
```

```
                ilo_age |   .1877888     .10043     1.87   0.062    -.0091304    .3847079
                        |
c.ilo_age#c.ilo_age |  -.0023906    .001183    -2.02   0.043    -.0047102   -.0000709
```

1

1 yes: −.0024, p = 0.043

QUICK CHECK

You change the base region with ib7. What happens to R-squared?

A. it rises

B. it falls

C. nothing: same model, other comparisons

ANSWER

You change the base region with ib7. What happens to R-squared?

A it rises

B it falls

C nothing: same model, other comparisons

Logs R02 and R03: .0227 both times.

Part 2 on one page

CHEAT CARD · KEEP THIS ONE

`regress y x if emp == 1`	linear regression
`i.region`	one coefficient per category
`ib7.region`	category 7 as the base
`c.age##c.age`	age and age squared
`predict yhat if e(sample)`	fitted values
`predict res, residuals`	residuals
`display e(N) e(r2)`	stored results

Practice on your own

PRACTICE SET · practice_D4.do, PART 2

CORE	P1	Regress usual hours on sex only. Type the coefficient for women.
CORE	P2	Add age group. Type R-squared, four decimals.
CORE	P3	Make Central 2 the base region. Type the coefficient for North.
EXTENSION	P4	Add the institutional sector. Type the coefficient for private.
CHALLENGE	P5	How many employed persons have a residual above 40?

Each exercise has a CHECK line: silent means correct. Solutions are released at the end of the day.

You can fit and read a regression with categorical characteristics.

lost? open **catchup_D4_S2.do**



3

DAY 4 · PART 3

Can we trust the model?

THE PROBLEM

The education coefficients looked strange: −8 hours each, SE 6.6, none significant

WHAT WE SAW

Four levels all about 8 hours below the base. A constant of 49 hours.

WHAT WE NEED

To know whether the model is sound, and which standard errors to report.

WHAT CAN GO WRONG

Unequal spread, overlapping characteristics, a few rows that steer the fit, clustered households.

Four checks, then the standard error NBS reports.

Part 3: checking a model

AFTER THIS PART YOU CAN

- read a residual plot and test for unequal spread
- spot overlapping characteristics with `estat vif`
- find the rows that steer the fit (leverage)
- choose between robust, clustered and `svy:` standard errors

Residuals against fitted values:



YOUR TURN · TASK 18

Draw the residuals

DO

Type:

```
regress ilo_job1_how_usual i.ilo_sex i.agegrp
i.region ///
    i.ilo_edu_aggregate if emp == 1
rvfplot, yline(0)
```

YOU SHOULD SEE

A cloud of points around the zero line, with a few far to the right.

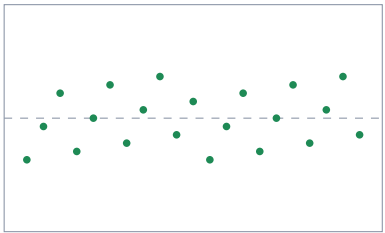
IF NOT

Read the residual plot

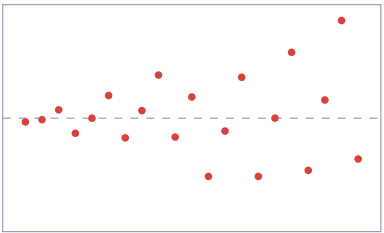


1. three points far to the right: fitted near 50 hours
2. residuals up to +78: people reporting 80 to 120 hours
3. rows of dots: hours are reported in round numbers

What a good and a bad residual plot look like



even band: constant spread



funnel: spread grows

A funnel means the standard errors from plain `regress` are not reliable. The fix is a different standard error, not a different model.

A test for unequal spread:

```
estat hettest
```

Breusch-Pagan
test, after regress

YOUR TURN · TASK 19

Test the spread

DO

Type:

```
estat hettest
```

YOU SHOULD SEE

A chi-squared statistic and its p-value.

IF NOT

Read the test

```
. estat hettest

Breusch-Pagan/Cook-Weisberg test for heteroskedasticity
Assumption: Normal error terms
Variable: Fitted values of ilo_job1_how_usual

H0: Constant variance (1)

    chi2(1) =    8.31
Prob > chi2 = 0.0039 (2)
```

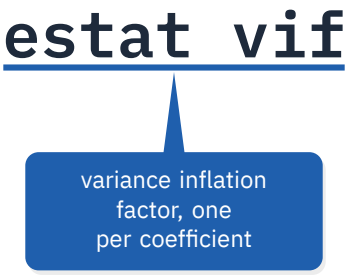
1. the hypothesis: constant spread
2. p = 0.004: rejected; the spread is not constant

The spread is not constant.

The coefficients are still fine. Their plain standard errors are not.

Use robust, clustered or `svy:` standard errors (this part).

Do two characteristics tell the same story?



YOUR TURN · TASK 20

Check the overlap

DO

Type:

```
estat vif
```

YOU SHOULD SEE

One VIF per coefficient and the mean VIF.

IF NOT

EXTRA Which variable has the largest VIF?

Read the VIFs

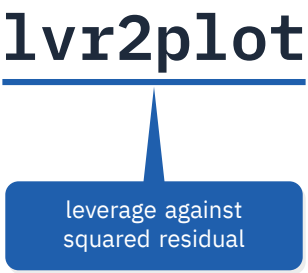
```
. estat vif

    Variable |       VIF       1/VIF
-------------+----------------------
   2.ilo_sex |      1.01    0.989622
      agegrp |
          2  |      2.51    0.399162
          3  |      3.02    0.330916
          4  |      2.62    0.382004
      region |
          2  |      1.62    0.617351
          3  |      1.53    0.653018
          4  |      1.87    0.534792
          5  |      1.64    0.610843
          6  |      1.41    0.709323
          7  |      1.28    0.782434
ilo_edu_ag~e |
          2  |     81.44    0.012279
          3  |    162.22    0.006165
          4  |     81.60    0.012255
          5  |     25.72    0.038883
-------------+----------------------
    Mean VIF |     26.39
```

1. sex, age group, region: below 3, fine
2. education: up to 162
3. mean 26.4

A huge VIF on every level of one variable often means an almost empty base category.

Rows that steer the fit:



YOUR TURN · TASK 21

Draw the leverage plot

DO

Type:

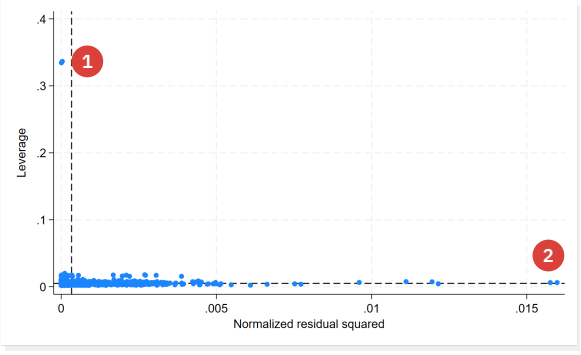
```
lvr2plot
```

YOU SHOULD SEE

Most points near zero, two or three high on the left.

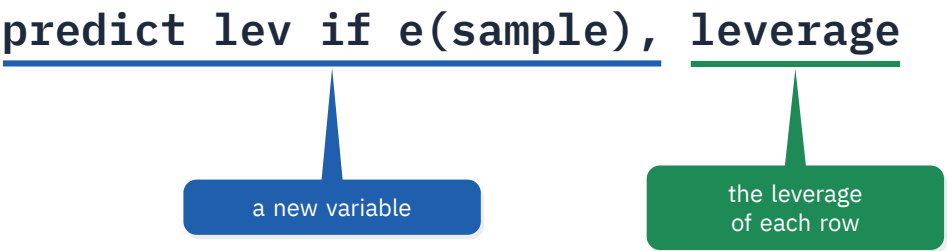
IF NOT

Read the leverage plot



1. leverage .33: a hundred times the average .005
2. large residuals, ordinary leverage: not influential

Who are they?


predict lev if e(sample), leverage
a new variable
the leverage
of each row

YOUR TURN · TASK 22

Find the high-leverage rows

DO

Type:

```
regress ilo_job1_how_usual i.ilo_sex i.agegrp
i.region ///
    i.ilo_edu_aggregate if emp == 1
predict lev if e(sample), leverage
list unique_ind ilo_edu_aggregate
ilo_job1_how_usual lev ///
    if lev > .1 & !missing(lev)
```

YOU SHOULD SEE

Three rows.

IF NOT

Three people

```
. list unique_ind ilo_edu_aggregate ilo_job1_how_usual lev if lev > .1 & !missing(lev)

     +-------------------------------------------------------+
     | uniqu~nd     ilo_edu_aggregate   ilo_jo..        lev |
     |-------------------------------------------------------|
2462. |  2218141   1 - Less than basic         50   .3364896 | (1)
4481. |  4140231   1 - Less than basic         48   .3348066 |
4529. |  4140232   1 - Less than basic         48   .3345464 |
     +-------------------------------------------------------+
```

(1) all three: ”Less than basic”, working 48 to 50 hours

YOUR TURN · TASK 23

How many employed in each education level?

DO

Type:

```
tabulate ilo_edu_aggregate if emp == 1 &
!missing(ilo_job1_how_usual)
```

YOU SHOULD SEE

Five rows; the first is tiny.

IF NOT

The base had three people

```
. tabulate ilo_edu_aggregate if emp == 1 & !missing(ilo_job1_how_usual)

Education (Aggregate |
              level) |      Freq.     Percent        Cum.
---------------------+-----------------------------------
 1 - Less than basic |          3        0.10        0.10
           2 - Basic |        264        8.80        8.90
    3 - Intermediate |      2,391       79.73       88.63
        4 - Advanced |        265        8.84       97.47
5 - Level not stated |         76        2.53      100.00
---------------------+-----------------------------------
               Total |      2,999      100.00
```

1. 3 employed persons: every education coefficient was a comparison with them
2. 2,391: a solid base

The fix: a large base category

`ib3.ilo_edu_aggregate`

Intermediate
as the base

YOUR TURN · TASK 24

Refit with Intermediate as the base

DO

Type:

```
regress ilo_job1_how_usual i.ilo_sex i.agegrp
i.region ///
    ib3.ilo_edu_aggregate if emp == 1
estat vif
```

YOU SHOULD SEE

Education rows near zero with small SEs; every VIF below 3.1.

IF NOT

The VIFs, before and after

BASE: 3 PEOPLE

```
ilo_edu_ag~e |
           2 |      81.44    0.012279
           3 |     162.22    0.006165
           4 |      81.60    0.012255
           5 |      25.72    0.038883
-------------+---------------------
    Mean VIF |      26.39
```

BASE: 2,391 PEOPLE

```
ilo_edu_ag~e |
           1 |       1.00    0.997073
           2 |       1.17    0.858329
           4 |       1.03    0.970738
           5 |       1.03    0.971151
    Mean VIF |       1.62
```

1 up to 162 2 mean VIF 1.62. The overlap was never real.

Read education again

```
. regress ilo_job1_how_usual i.ilo_sex i.agegrp i.region ib3.ilo_edu_aggregate if emp == 1
    Residual |  385959.489     2,984  129.342992   R-squared       =    0.0237 (1)
    ilo_edu_aggregate |
 1 - Less than basic  |   8.195958   6.579073     1.25   0.213     -4.70402    21.09594
           2 - Basic  |   .4796628   .7911379     0.61   0.544    -1.071568    2.030894 (2)
        4 - Advanced  |  -.7238596   .7426538    -0.97   0.330    -2.180025    .7323057 (3)
5 - Level not stated  |   .3309336   1.340896     0.25   0.805    -2.298241    2.960109
                      |
                _cons |   41.23025   .8373421    49.24   0.000     39.58842    42.87208
------------------------------------------------------------------------------------------
```

1. R-squared .0237: the same fit
2. Basic against Intermediate: +.5 hours, p = 0.54
3. Advanced: −.7, p = 0.33. Education does not change hours much

The base fixed the VIF, not the leverage: three people still sit far to the right. In the lab: merge them into Basic.

Standard errors that allow unequal spread:



YOUR TURN · TASK 25

Robust standard errors

DO

Type:

```
regress ilo_job1_how_usual i.ilo_sex i.agegrp
i.region ///
    i.ilo_edu_aggregate if emp == 1, vce(robust)
```

YOU SHOULD SEE

The same coefficients; the column now says ”Robust std. err.”.

IF NOT

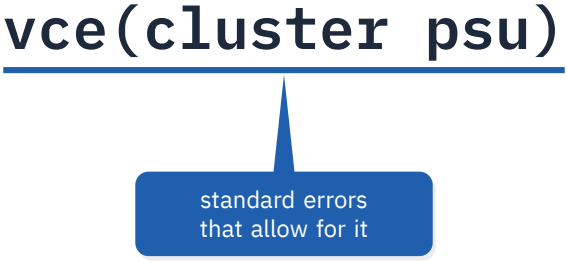
EXTRA Did the coefficient for women change?

Read the robust model

```
. regress ilo_job1_how_usual i.ilo_sex i.agegrp i.region i.ilo_edu_aggregate if emp == 1, vce(robust)
Linear regression                               Number of obs     =      2,999
                                                F(14, 2984)       =       7.45
                                                Prob > F          =     0.0000
                                                R-squared         =     0.0237
------------------------------------------------------------------------------
                             |               Robust
   ilo_job1_how_usual | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
----------------------+-------------------------------------------------------
             ilo_sex |
         2 - Female  |  -2.927601   .4226992    -6.93   0.000    -3.756412   -2.098789
```

1. the column header tells you which SE this is
2. women: −2.93, SE .423 (plain .418)

People in the same PSU are alike:



YOUR TURN · TASK 26

Clustered standard errors

DO

Type:

```
regress ilo_job1_how_usual i.ilo_sex i.agegrp
i.region ///
    i.ilo_edu_aggregate if emp == 1, vce(cluster
psu)
```

YOU SHOULD SEE

A note: standard errors adjusted for 146 clusters.

IF NOT

Read the clustered model

```
. regress ilo_job1_how_usual i.ilo_sex i.agegrp i.region i.ilo_edu_aggregate if emp == 1, vce(cluster psu)
Linear regression                               Number of obs     =      2,999
                                                F(14, 145)        =       6.26
                                              (Std. err. adjusted for 146 clusters in psu)
                     |               Robust
   ilo_job1_how_usual | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
----------------------+------------------------------------------------------------------
             ilo_sex |
          2 - Female |  -2.927601   .4360237    -6.71   0.000    -3.789384   -2.065818
```

1. degrees of freedom: 145 = clusters − 1
2. 146 PSUs
3. women: SE .436

Weights, strata and clusters together:



YOUR TURN · TASK 27

The design-based regression

DO

Type:

```
svy, subpop(emp): regress ilo_job1_how_usual
i.ilo_sex ///
    i.agegrp i.region i.ilo_edu_aggregate
```

YOU SHOULD SEE

The survey header (26 strata, 146 PSUs, 2,999 in the subpopulation) and a new coefficient table.

IF NOT

Read the design-based model

```
. svy, subpop(emp): regress ilo_job1_how_usual i.ilo_sex i.agegrp i.region i.ilo_edu_aggregate
Number of strata =  26                        Number of obs    =      5,875
Number of PSUs   = 146                        Population size = 101,093.06
Subpop. no. PSUs = 146                        Subpop. no. obs =      2,999   (1)
                                              Subpop. size    = 52,160.781
                                              Design df       =        120   (2)
                                              F(14, 107)      =       6.05
                                              Prob > F        =     0.0000
                                              R-squared       =     0.0284
                     |             Linearized
   ilo_job1_how_usual | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
---------------------+------------------------------------------------------------------
              ilo_sex |
          2 - Female  |  -3.111163   .4639096    -6.71   0.000    -4.029672   -2.192654   (3)
```

1. 2,999 employed: 52,161 people
2. design df 120 = 146 − 26
3. women: −3.11, SE .464

YOUR TURN · TASK 28

Three standard errors side by side

DO

Type:

```
quietly regress ilo_job1_how_usual i.ilo_sex
i.agegrp ///
    i.region i.ilo_edu_aggregate if emp == 1
estimates store plain
(same with vce(cluster psu): store clus)
(same with svy, subpop(emp): store design)
estimates table plain clus design, ///
    keep(2.ilo_sex) b(%7.3f) se(%7.3f)
```

YOU SHOULD SEE

One small table: three coefficients, three SEs.

IF NOT

Same model, three standard errors

```
. estimates table plain clus design, keep(2.ilo_sex) b(%7.3f) se(%7.3f)

-------------------------------------------
    Variable |  plain     clus     design
-------------+-----------------------------
     ilo_sex |
 2 - Female  |  -2.928    -2.928    -3.111
             |   0.418     0.436     0.464
-------------------------------------------
                              Legend: b/se
```

1. plain and clustered: −2.93; design: −3.11
2. SE .418, .436, .464: each adds a part of the truth

For NBS figures, the `svy:` column is the one to report.

COMMON ERROR

A plain-regression check after svy

```
. estat hettest
command estat hettest is not supported after svy estimation
```

WHY

`estat hettest` assumes a plain regression. After `svy:` it is not defined.

FIX

Run the checks on the plain model; report the `svy:` model.

COMMON ERROR

One letter swapped

```
. regress ilo_job1_how_usual i.ilo_sex if emp == 1, vce(cluster pus)
variable pus not found
```

WHY

There is no variable `pus`.

FIX

Type `psu`. Tab-complete variable names.

ON YOUR OWN · E

Do the Advanced work fewer hours?

THE QUESTION

With Intermediate as the base, do employed persons with Advanced education work fewer hours? Is the difference significant?

HINT

`ib3.ilo_edu_aggregate`.

ON YOUR OWN · E · ANSWER

```
regress ilo_job1_how_usual i.ilo_sex i.agegrp ///
i.region ib3.ilo_edu_aggregate if emp == 1
```

```
ilo_edu_aggregate |
1 - Less than basic |   8.195958   6.579073     1.25   0.213     -4.70402    21.09594
          2 - Basic |   .4796628   .7911379     0.61   0.544    -1.071568    2.030894
       4 - Advanced |  -.7238596   .7426538    -0.97   0.330    -2.180025    .7323057
5 - Level not stated |   .3309336   1.340896     0.25   0.805    -2.298241    2.960109
```

1

1 0.72 fewer, p = 0.33: no

ON YOUR OWN · F

How many rows steer the fit?

THE QUESTION

After the model with `i.ilo_edu_aggregate`, how many employed persons have a leverage above .1?

HINT

`predict ..., leverage`, then `count if`.

ON YOUR OWN · F · ANSWER

```
predict lev if e(sample), leverage
summarize lev, detail
```

```
. summarize lev, detail

                            Leverage
-------------------------------------------------------------
      Percentiles      Smallest
 1%     .0020704       .0020704
 5%     .0020704       .0020704
10%     .0027786       .0020704       Obs               2,999
25%     .0029193       .0020704       Sum of wgt.       2,999

50%      .003947                      Mean           .0050017
                        Largest       Std. dev.      .0107622
75%     .0058266       .0199744
90%     .0071803       .3345464       Variance       .0001158
95%     .0085391       .3348066       Skewness       28.95656
99%      .016434       .3364896       Kurtosis       887.9673  1
```

1. the three largest are above .33; all others below .02: three

QUICK CHECK

Every education level has a VIF above 25. The most likely cause?

A. education and age measure the same thing

B. the base category is almost empty

C. the data are wrong

ANSWER

Every education level has a VIF above 25. The most likely cause?

A education and age measure the same thing

B the base category is almost empty

C the data are wrong

Three people in the base (log G14). With ib3. every VIF is below 3.1 (log G15).

Part 3 on one page

CHEAT CARD · KEEP THIS ONE

`rvfplot, yline(0)`	residuals against fitted
`estat hettest`	unequal spread? (plain regress)
`estat vif`	overlapping characteristics
`lvr2plot`	rows that steer the fit
`predict lev, leverage`	leverage per row
`regress ..., vce(cluster psu)`	clustered SE
`svy, subpop(emp): regress ...`	the SE NBS reports

Practice on your own

PRACTICE SET · practice_D4.do, PART 3

CORE P1 Hours on sex and age group: `estat hettest`. Type the p-value.

CORE P2 Same model, `vce(cluster psu)`: type the SE for women.

CORE P3 Same model with `svy, subpop(emp):` type the coefficient for women.

EXTENSION P4 Merge ”Less than basic” into ”Basic” and refit: type the largest leverage.

CHALLENGE P5 How many employed report more than 80 usual hours?

Each exercise has a CHECK line: silent means correct. Solutions are released at the end of the day.

You can check a model and report it with the right standard errors.

lost? open **catchup_D4_S3.do**

Lab 1 · Hours worked

THE SITUATION

The Minister's office asked whether women in employment work fewer hours, all else equal, and whether the public sector differs from the private sector.

OPEN

`lab_D4_1.do` in `C:\NBS_Stata\day4`.

THE TASKS

CORE Tasks 1 to 5: design-based gap, model, checks, `svy:` model

EXTENSION Task 6: add the institutional sector

CHALLENGE Task 7: does the gap differ between sectors?

CORRECT WHEN

the file runs without red text and every CHECK is silent.

The lab file: lab_D4_1.do

```
* TASK 2 (Core)  The design-based gap in usual hours
*   Estimate mean usual hours of the employed by sex with svy, then the
*   gap men minus women with lincom.
* ------------------------------------------------------------------------
* >>> YOUR CODE



* CHECK Task 2 (silent = correct)
assert abs(r(estimate) - 3.048675) < 1e-5
```

Type under >>> `YOUR CODE`. A silent CHECK means correct.

Green note when your last check is silent. Then the Challenge task.



4

DAY 4 · PART 4

Explaining a yes or no

THE PROBLEM

Who is more likely to be unemployed: the young, women, people on Praslin?

WHAT WE HAVE

A 0/1 flag: unemployed or not, for the 3,126 in the labour force.

WHAT WE NEED

The chance of being unemployed for each group, all else equal.

WHY NOT REGRESS

A straight line can predict below 0 or above 1. A chance cannot.

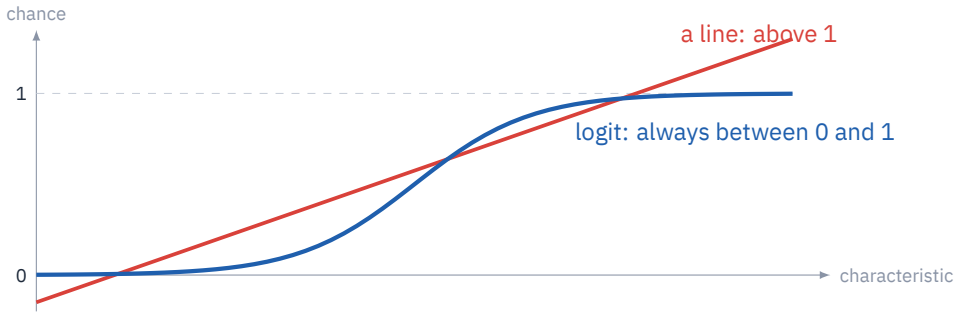
Logit keeps every prediction between 0 and 1.

Part 4: models for a yes or no

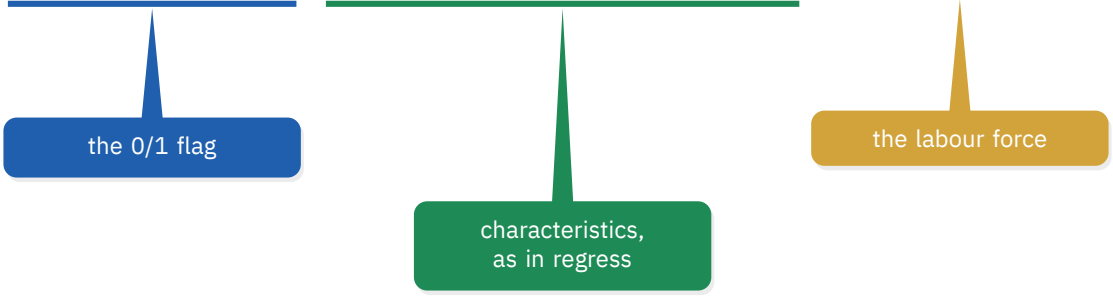
AFTER THIS PART YOU CAN

- fit `logit` and say why its coefficients are not percentage points
- turn the model into predicted rates with `margins`
- give a difference in percentage points with `dydx()`
- chart the result with `marginsplot`, design-based

A line escapes, an S-curve stays inside



A model for a 0/1 outcome:



YOUR TURN · TASK 29

Fit the logit

DO

Type:

```
logit unemp i.ilo_sex i.agegrp if lf == 1
```

YOU SHOULD SEE

A few ”Iteration” lines, then a table like regress.

IF NOT

Read the logit

```
. logit unemp i.ilo_sex i.agegrp if lf == 1

Logistic regression                                     Number of obs =  3,126
                                                        LR chi2(4)    = 110.23
                                                        Prob > chi2   = 0.0000
Log likelihood = -434.27195                             Pseudo R2     = 0.1126
------------------------------------------------------------------------------
       unemp | Coefficient  Std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
     ilo_sex |
 2 - Female  |   .0151041   .1963335     0.08   0.939    -.3697025    .3999107
             |
      agegrp |
       25-34 |  -1.719609   .2609406    -6.59   0.000    -2.231044   -1.208175
       35-54 |  -2.230147   .2399429    -9.29   0.000    -2.700427   -1.759868
         55+ |  -2.655709   .3844162    -6.91   0.000    -3.409151   -1.902267
             |
       _cons |  -1.702678   .1767841    -9.63   0.000    -2.049169   -1.356188
```

1. 3,126 in the labour force
2. women: +.015, p = 0.94: no difference
3. every older group: negative, p = 0.000

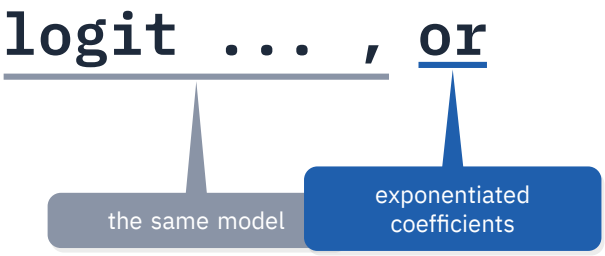
WARNING

A logit coefficient is not a change in percentage points.

It is a change in the log of the odds. Nobody reads a release in log-odds.

Read logit through `margins`.

Odds ratios instead:



YOUR TURN · TASK 30

Odds ratios

DO

Type:

```
logit unemp i.ilo_sex i.agegrp if lf == 1, or
```

YOU SHOULD SEE

The column header says ”Odds ratio”.

IF NOT

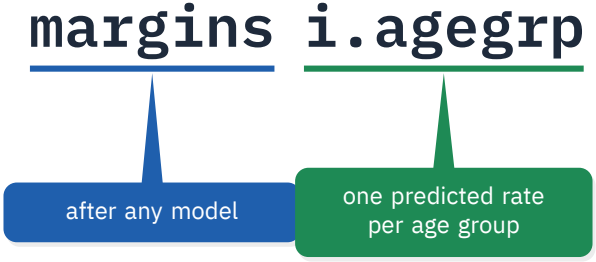
Read the odds ratios

```
. logit unemp i.ilo_sex i.agegrp if lf == 1, or
------------------------------------------------------------------------------
       unemp | Odds ratio   Std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
     ilo_sex |
 2 - Female  |   1.015219   .1993214     0.08   0.939     .6909398    1.491691
             |
      agegrp |
       25-34 |   .1791361   .0467439    -6.59   0.000     .1074163    .2987419   1
       35-54 |   .1075126   .0257969    -9.29   0.000     .0671768    .1720676
         55+ |    .070249   .0270049    -6.91   0.000     .0330693     .14923
             |
       _cons |   .1821949   .0322092    -9.63   0.000      .128842    .2576411   2
------------------------------------------------------------------------------
Note: _cons estimates baseline odds.
```

1. the odds for 25-34 are .18 times those of 15-24

2. baseline odds for men 15-24: .18

Predicted rates for each group:



YOUR TURN · TASK 31

Predicted rates by age group

DO

Type:

```
logit unemp i.ilo_sex i.agegrp if lf == 1
margins i.agegrp
```

YOU SHOULD SEE

Four rows: a predicted unemployment rate per age group, with interval.

IF NOT

EXTRA Run `margins i.ilo_sex` too.

Read the predicted rates

```
. margins i.agegrp
------------------------------------------------------------------------------
             |            Delta-method
             |     Margin   std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
      agegrp |
      15-24  |   .1551479   .0190585     8.14   0.000     .1177939    .1925019   1
      25-34  |   .0318489   .0066827     4.77   0.000     .0187511    .0449467
      35-54  |   .0193614   .0036235     5.34   0.000     .0122595    .0264633
        55+  |   .0127363   .0044743     2.85   0.004     .0039668    .0215058   2
------------------------------------------------------------------------------
```

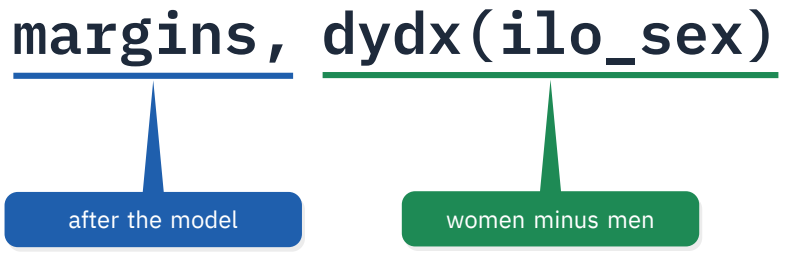
1. 15-24: 15.5%
2. 55 and over: 1.3%

And by sex

```
. margins i.ilo_sex
------------------------------------------------------------------------------
             |            Delta-method
             |     Margin   std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
     ilo_sex |
   1 - Male  |   .0362099   .0046667     7.76   0.000     .0270633    .0453566   1
 2 - Female  |   .0367122   .0045621     8.05   0.000     .0277707    .0456537   2
------------------------------------------------------------------------------
```

1. men 3.6%
2. women 3.7%: almost the same

A difference in percentage points:



YOUR TURN · TASK 32

Women against men, in points

DO

Type:

```
margins, dydx(ilo_sex)
```

YOU SHOULD SEE

One row: the difference, with SE and interval.

IF NOT

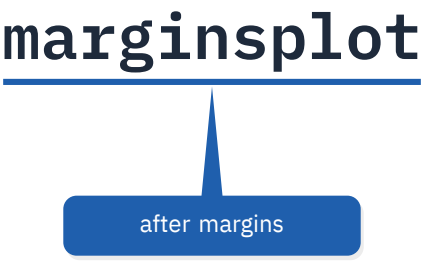
Read the difference

```
. margins, dydx(ilo_sex)
                |            Delta-method
                |      dy/dx   std. err.      z    P>|z|     [95% conf. interval]
----------------+------------------------------------------------------------------
     ilo_sex |
 2 - Female  |   .0005022   .0065272     0.08   0.939    -.0122909    .0132954
------------------------------------------------------------------------------
Note: dy/dx for factor levels is the discrete change from the base level.
```

1

1 +.05 points, from −1.2 to +1.3: no difference

Chart the predicted rates:



YOUR TURN · TASK 33

Chart by age group

DO

Type:

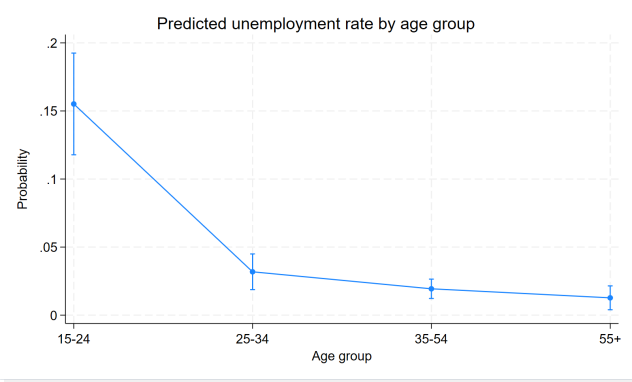
```
margins i.agegrp
marginsplot, title("Predicted unemployment rate by
age group") ///
    ytitle("Probability")
```

YOU SHOULD SEE

A line with four points and their intervals.

IF NOT

The chart, unweighted



Youth 15.5%, then a steep fall. Unweighted: the next step adds the design.

The design-based logit:

```
svy, subpop(lf): logit unemp ...
```

the design and the labour force

the same model

Education breaks the logit

```
. svy, subpop(lf): logit unemp i.ilo_sex i.agegrp i.ilo_edu_aggregate i.region
note: 1b.ilo_edu_aggregate != 0 predicts failure perfectly;
      1b.ilo_edu_aggregate omitted and 3 obs not used.
note: 5.ilo_edu_aggregate != 0 predicts failure perfectly;
      5.ilo_edu_aggregate omitted and 76 obs not used.
note: 4.ilo_edu_aggregate omitted because of collinearity.
    ilo_edu_aggregate |
 1 - Less than basic  |          0  (empty)
           2 - Basic  |   .7977621   .6949514     1.15   0.253    -.5781934    2.173718
    3 - Intermediate  |   1.165527    .491928     2.37   0.019      .191544     2.13951
        4 - Advanced  |          0  (omitted)
5 - Level not stated  |          0  (empty)
. margins i.agegrp
missing predicted values encountered within the estimation sample
```

1. a level with no unemployed person: dropped
2. then another, for collinearity
3. `margins` cannot predict for the dropped rows

YOUR TURN · TASK 34

Why?

DO

Type:

```
tabulate ilo_edu_aggregate unemp if lf == 1
```

YOU SHOULD SEE

Two levels with no unemployed person.

IF NOT

No unemployed, no estimate

```
. tabulate ilo_edu_aggregate unemp if lf == 1

Education (Aggregate |         unemp
              level) |         0          1 |     Total
---------------------+----------------------+----------
 1 - Less than basic |         3          0 |         3 (1)
           2 - Basic |       265          7 |       272
    3 - Intermediate |     2,401        101 |     2,502
        4 - Advanced |       267          6 |       273
5 - Level not stated |        76          0 |        76 (2)
---------------------+----------------------+----------
               Total |     3,012        114 |     3,126
```

1. 3 in the labour force, 0 unemployed
2. 76, 0 unemployed

Leave education out of the unemployment model.

With 114 unemployed in the sample, a characteristic with empty cells breaks the model.

Sex, age group and region.

YOUR TURN · TASK 35

The design-based model we keep

DO

Type:

```
svy, subpop(lf): logit unemp i.ilo_sex i.agegrp
i.region
```

YOU SHOULD SEE

The survey header and a table: sex, three age groups, six regions.

IF NOT

Read the design-based logit

```
. svy, subpop(lf): logit unemp i.ilo_sex i.agegrp i.region
Subpop. no. PSUs = 146                          Subpop. no. obs =      3,126 (1)
                                                Subpop. size    = 54,224.639
                                                Design df       =        120
                ilo_sex |
          2 - Female    |  -.1113344   .2793506    -0.40   0.691    -.6644291    .4417603 (2)
                        |
                agegrp  |
                25-34   |  -1.531704   .3107664    -4.93   0.000       -2.147   -.9164086
                35-54   |  -1.981503   .2775866    -7.14   0.000    -2.531105     -1.4319
                  55+   |  -2.412999    .402621    -5.99   0.000    -3.210161   -1.615837 (3)
6- Praslin & LaDigue    |  -1.446413   .5728247    -2.53   0.013    -2.580566   -.3122597 (4)
```

1. 3,126 in the labour force
2. women: no difference, p = 0.69
3. every older group lower, p = 0.000
4. Praslin & La Digue lower, p = 0.013

YOUR TURN · TASK 36

Design-based predicted rates

DO

Type:

```
margins i.agegrp
marginsplot, title("Predicted unemployment rate by
age group") ///
    ytitle("Probability") ///
    note("Source: NBS, LFS Q1 2026 (teaching
file), design-based")
```

YOU SHOULD SEE

Four predicted rates, then the chart with a source note.

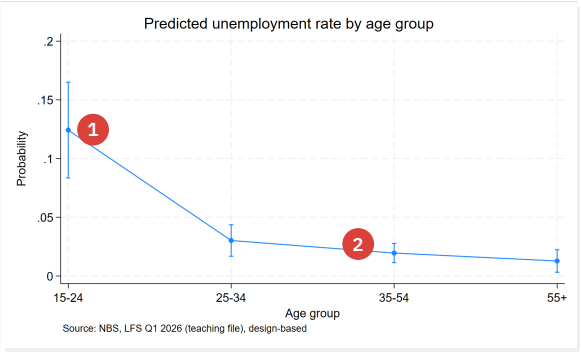
IF NOT

Read the design-based rates

```
. margins i.agegrp
------------------------------------------------------------------------------
             |            Delta-method
             |     Margin   std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
      agegrp |
       15-24 |   .1242552    .020608     6.03   0.000     .0834528    .1650576
       25-34 |   .0302085     .006771    4.46   0.000     .0168023    .0436147
       35-54 |   .0195111    .0041259    4.73   0.000      .011342    .0276801
         55+ |   .0127737    .0048513    2.63   0.010     .0031685    .0223789
```

1. 15-24: 12.4%, from 8.3 to 16.5
2. 55 and over: 1.3%

The chart for the note



1. young people: about one in eight
2. from 25 on: below 3%

Unweighted and design-based

UNWEIGHTED

```
15-24 |   .1551479   .0190585     8.14   0.000     .1177939    .1925019
```

1

DESIGN-BASED

```
15-24 |   .1242552    .020608     6.03   0.000     .0834528    .1650576
```

2

1 15.5% 2 12.4%, with region in the model. The weights move the estimate by three points.

YOUR TURN · TASK 37

Differences in points, design-based

DO

Type:

```
margins, dydx(ilo_sex agegrp)
```

YOU SHOULD SEE

Four rows: women, and each older age group against 15-24.

IF NOT

Read the differences

```
. margins, dydx(ilo_sex agegrp)
                |            Delta-method
                |      dy/dx   std. err.      t    P>|t|     [95% conf. interval]
----------------+------------------------------------------------------------------
     ilo_sex    |
 2 - Female     |  -.0035903   .0090017    -0.40   0.691    -.0214129    .0142324   1
                |
      agegrp    |
      25-34     |  -.0940467   .0220246    -4.27   0.000    -.1376538   -.0504396   2
      35-54     |  -.1047441    .020653    -5.07   0.000    -.1456356   -.0638527
        55+     |  -.1114815   .0203651    -5.47   0.000     -.151803   -.0711599
```

1. women: −.4 points, p = 0.69
2. 25-34: 9.4 points below 15-24

COMMON ERROR

An outcome that is not 0/1

```
. logit ilo_lfs19 i.ilo_sex if lf == 1
outcome does not vary; remember:
                                0 = negative outcome,
        all other nonmissing values = positive outcome
```

WHY

`logit` reads 0 as no and anything else as yes. In the labour force `ilo_lfs19` is 1 or 2: never 0, so it never varies.

FIX

Build a 0/1 flag first: `unemp = (ilo_lfs19 == 2)`.

COMMON ERROR

margins for a characteristic not in the model

```
. margins i.region
factor region not found in list of covariates
```

WHY

The model has sex and age group. Region is not in it.

FIX

Add `i.region` to the model, then ask.

COMMON ERROR

margins for a number

```
. margins ilo_age
factor ilo_age not found in list of covariates
```

WHY

`margins ilo_age` wants a factor variable, and age is not in this model.

FIX

For a continuous variable use `margins, at(ilo_age = (20 40 60))` after a model that has it.

Probit: another S-curve

```
svy, subpop(lf): probit
```

same syntax as logit

Logit and probit agree

LOGIT

```
15-24  |   .1242552   .020608     6.03   0.000     .0834528    .1650576
25-34  |   .0302085   .006771     4.46   0.000     .0168023    .0436147
```

PROBIT

```
15-24  |   .1240758   .0209528     5.92   0.000     .0825907    .1655609
25-34  |  -.7359384   .1480376    -4.97   0.000    -1.029043   -.4428343
```

1 12.43% 2 12.41%. Report whichever your office uses; read it through margins.

ON YOUR OWN · G

Young women

THE QUESTION

From the design-based model with sex and age group, what is the predicted unemployment rate of women aged 15-24?

HINT

```
margins ilo_sex#agegrp.
```

ON YOUR OWN · G · ANSWER

```
svy, subpop(lf): logit unemp i.ilo_sex i.agegrp
margins ilo_sex#agegrp
```

ilo_sex#agegrp	Margin	std. err.	t	P>|t|	[95% conf.	interval]
1 - Male#15-24	.1360265	.0283177	4.80	0.000	.0799594	.1920937
1 - Male#25-34	.0319939	.0085077	3.76	0.000	.0151492	.0488386
1 - Male#35-54	.0208359	.0053545	3.89	0.000	.0102343	.0314374
1 - Male#55+	.0140227	.0052301	2.68	0.008	.0036675	.0243778
2 - Female#15-24	.1210469	.0235011	5.15	0.000	.0745163	.1675775
2 - Female#25-34	.028098	.0072638	3.87	0.000	.0137162	.0424798
2 - Female#35-54	.0182731	.0043449	4.21	0.000	.0096706	.0268756
2 - Female#55+	.0122874	.0051325	2.39	0.018	.0021254	.0224493

1

1 12.1%, from 7.5 to 16.8

ON YOUR OWN · H

How much lower from 25?

THE QUESTION

Design-based: by how many percentage points is the 25-34 rate below the 15-24 rate? Give the interval.

HINT

```
margins, dydx(agegrp).
```

ON YOUR OWN · H · ANSWER

```
svy, subpop(lf): logit unemp i.ilo_sex ///
i.agegrp i.region
margins, dydx(ilo_sex agegrp)
```

```
             |            Delta-method
             |      dy/dx   std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
     ilo_sex |
 2 - Female  |  -.0035903   .0090017    -0.40   0.691    -.0214129    .0142324
             |
      agegrp |
       25-34 |  -.0940467   .0220246    -4.27   0.000    -.1376538   -.0504396
       35-54 |  -.1047441    .020653    -5.07   0.000    -.1456356   -.0638527
         55+ |  -.1114815   .0203651    -5.47   0.000     -.151803   -.0711599
```

1

1 9.4 points lower, 5.0 to 13.8

One in eight young people in the labour force is unemployed: four times the rate from age 25.

QUICK CHECK

The logit coefficient for 25-34 is −1.53. What does it mean?

A. the rate is 1.53 points lower

B. lower log-odds: read it with margins

C. the rate is 153% lower

ANSWER

The logit coefficient for 25-34 is −1.53. What does it mean?

A the rate is 1.53 points lower

B lower log-odds: read it with margins

C the rate is 153% lower

margins, dydx() gives the difference in points: −9.4 (log L09).

Part 4 on one page

CHEAT CARD · KEEP THIS ONE

`logit flag i.x if lf == 1`	model for a 0/1 outcome
`logit ..., or`	odds ratios
`margins i.agegrp`	predicted rate per group
`margins, dydx(ilo_sex)`	difference in points
`marginsplot`	chart of the last margins
`svy, subpop(lf): logit ...`	design-based
`tabulate x flag`	check for empty cells first

Practice on your own

PRACTICE SET · practice_D4.do, PART 4

CORE	P1	Logit of unemployment on region only. Type the coefficient for Praslin.
CORE	P2	`margins i.region`: type the predicted rate for Perseverance.
CORE	P3	Design-based: type the predicted rate for women.
EXTENSION	P4	Odds ratio for 55+ in the design-based model.
CHALLENGE	P5	`margins, at()` after a model with `c.ilo_age`: rate at age 20.

Each exercise has a CHECK line: silent means correct. Solutions are released at the end of the day.

You can show who is more likely to be unemployed, in percentage points, with the design respected.

lost? open **catchup_D4_S4.do**

5

DAY 4 · PART 5

From output to a reader’s table

THE PROBLEM

The Director will not read a Stata results window

WHAT WE HAVE

Two logit models in the Results window, one after the other.

WHAT WE NEED

One table, models side by side, in Word, plus the chart and two sentences.

WHY NOT COPY-PASTE

Rerun with corrected data and every pasted number is out of date.

Store, tabulate, export: from the do-file, every time.

Part 5: presenting models

AFTER THIS PART YOU CAN

- keep several models with `estimates store`
- put them side by side with `etable`
- export the table to Word
- write a short note with `putdocx`

Keep a model:



YOUR TURN · TASK 38

Store two models

DO

Type:

```
logit unemp i.ilo_sex i.agegrp if lf == 1
estimates store m_plain
svy, subpop(lf): logit unemp i.ilo_sex i.agegrp
estimates store m_svy
```

YOU SHOULD SEE

Two model outputs; the store lines are silent.

IF NOT

Side by side:

```
etable, estimates(m_plain m_svy)
```

estimation table (Stata 17)

the stored models

YOUR TURN · TASK 39

One table, two models

DO

Type:

```
etable, estimates(m_plain m_svy)
```

YOU SHOULD SEE

Two columns, labels instead of variable names, SEs in brackets.

IF NOT

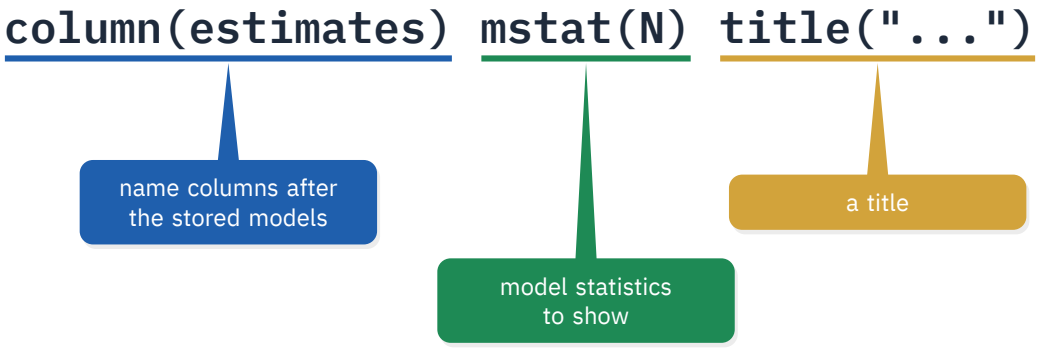
Read the table

```
. etable, estimates(m_plain m_svy)

---------------------------------------
                          unemp    unemp (1)
---------------------------------------
Sex
  2 - Female               0.015  -0.134 (2)
                          (0.196) (0.270) (3)
Age group
  25-34                   -1.720  -1.561
                          (0.261) (0.309)
  35-54                   -2.230  -2.001
                          (0.240) (0.269)
  55+                     -2.656  -2.404
                          (0.384) (0.398)
Intercept                 -1.703  -1.849
                          (0.177) (0.241)
Number of observations      3126    5888 (4)
---------------------------------------
```

1. both columns are called unemp: not helpful
2. value labels, not codes
3. SE in brackets
4. 3,126 and 5,888: svy counts every row

Column names and a title:



YOUR TURN · TASK 40

A titled table

DO

Type:

```
etable, estimates(m_plain m_svy) mstat(N) ///
    column(estimates) ///
    title("Unemployment, Q1 2026: logit
coefficients")
```

YOU SHOULD SEE

The same table with a title and columns m_plain and m_svy.

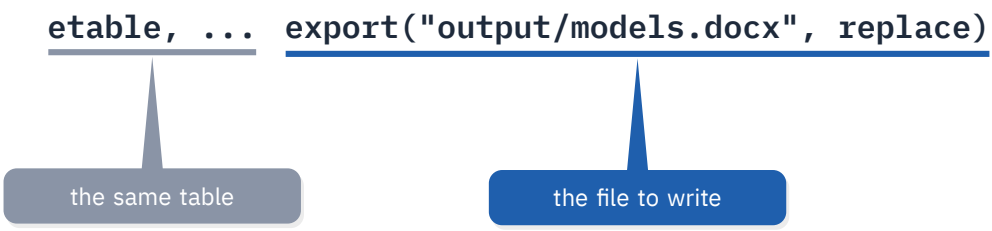
IF NOT

Read the titled table

```
Unemployment, Q1 2026: logit coefficients (1)
---------------------------------------
                        m_plain  m_svy (2)
---------------------------------------
Sex
  2 - Female             0.015  -0.134
                        (0.196) (0.270)
Age group
  25-34                 -1.720  -1.561
                        (0.261) (0.309)
  35-54                 -2.230  -2.001
                        (0.240) (0.269)
  55+                   -2.656  -2.404
                        (0.384) (0.398)
Intercept               -1.703  -1.849
                        (0.177) (0.241)
Number of observations    3126    5888
---------------------------------------
```

1. the title
2. column names from the stores

Straight into Word:



YOUR TURN · TASK 41

Export the table

DO

Type:

```
etable, estimates(m_plain m_svy) mstat(N) ///
    column(estimates) export("output/models.docx",
replace)
```

YOU SHOULD SEE

The table, then a line: collection ETable exported to file.

IF NOT

Close the file in Word before running again.

The export line

```
. etable, estimates(m_plain m_svy) mstat(N) column(estimates) ///
>     export("_trainer/capture/x_D4_models.docx", replace)
(collection ETable exported to file _trainer/capture/x_D4_models.docx)
```

1

1 written: open it from the output folder

The table in the document

	m_plain	m_svy
Sex		
2 - Female	0.015	-0.134
	(0.196)	(0.270)
Age group		
25-34	-1.720	-1.561
	(0.261)	(0.309)
35-54	-2.230	-2.001
	(0.240)	(0.269)
55+	-2.656	-2.404
	(0.384)	(0.398)
Intercept	-1.703	-1.849
	(0.177)	(0.241)
Number of observations	3126	5888

Rendered from `x_D4_models.docx`, the file written by the capture run. Edit fonts in Word; the numbers come from Stata.

COMMON ERROR

A model that was never stored

```
. etable, estimates(m_plain m_design)
estimation result m_design not found
```

WHY

The names in `estimates()` must match the `estimates store` names exactly.

FIX

`estimates dir` lists what is stored.

YOUR TURN · TASK 42

What is stored?

DO

Type:

```
estimates dir
```

YOU SHOULD SEE

One row per stored model.

IF NOT

The stored models

```
. estimates dir

-------------------------------------------------------------------------------
             |           Dependent  Number of
        Name | Command    variable     param.  Title
-------------+-----------------------------------------------------------------
       plain | regress   ~1_how_usual      19  Linear regression
        clus | regress   ~1_how_usual      19  Linear regression
      design | regress   ~1_how_usual      19  Survey: Linear regression
     m_plain | logit         unemp          7  Logistic regression
       m_svy | logit         unemp          7  Survey: Logistic regression
-------------------------------------------------------------------------------
```

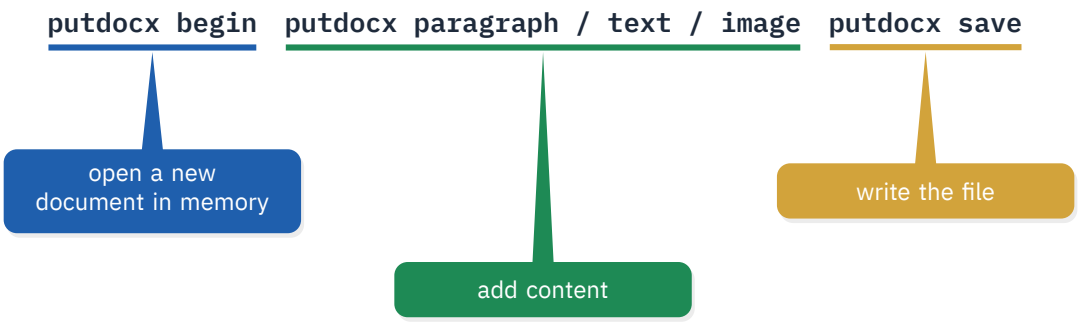
1. Part 3's three hours models
2. today's two logits

You will meet `esttab` in older NBS do-files

```
* user-written (SSC package estout): on the USB offline installer
esttab m_plain m_svy using "output/models.rtf", replace se
```

`esttab` was the standard before Stata 17. It needs the `estout` package, which needs installing. `etable` needs nothing: use it in new work.

A note in Word, from the do-file (Day 2):



A number inside a sentence

```
margins i.agegrp
matrix M = r(table)
putdocx begin
putdocx paragraph
putdocx text ("Young people aged 15-24: ")
putdocx text (100 * M[1, 1]), nformat(%4.1f)
putdocx text (" per cent.")
putdocx save "output/note_q1.docx", replace
```

- `putdocx text (expression)` with `nformat(%4.1f)`: no number typed by hand

YOUR TURN · TASK 43

Your first note

DO

After the design-based `margins` and `marginsplot`, type:

```
matrix M = r(table)
graph export "output/D4_margins.png", replace
width(1600)
putdocx clear
putdocx begin
putdocx paragraph, style(Title)
putdocx text ("Who is more likely to be
unemployed? Q1 2026")
putdocx paragraph
putdocx text (100 * M[1, 1]), nformat(%4.1f)
putdocx image "output/D4_margins.png"
putdocx save "output/note_q1.docx", replace
```

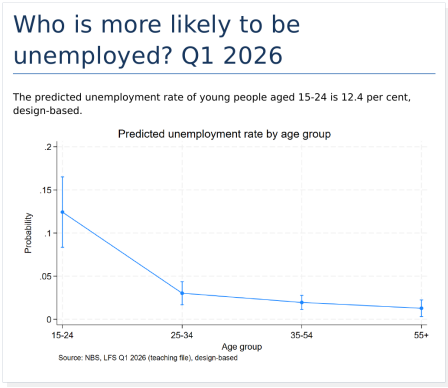
YOU SHOULD SEE

A line: successfully created. Open the file: a title, the youth rate and the chart.

IF NOT

Word says the file is in use? Close it in Word and run again.

The first page of the note



Page 1 of `note_q1.docx`: the title, the youth rate computed by Stata (12.4), the chart.

ON YOUR OWN · I

Hours models in one table

THE QUESTION

Put Part 3's three hours models (plain, clustered, design) in one `etable`, showing only the coefficient for women.

HINT

```
keep(2.ilo_sex).
```

ON YOUR OWN · I · ANSWER

```
etable, estimates(plain clus design) ///
keep(2.ilo_sex) mstat(N)
```

```
------------------------------------------------------------------------------
                    ilo_job1_how_usual ilo_job1_how_usual ilo_job1_how_usual
------------------------------------------------------------------------------
Sex
  2 - Female                    -2.928             -2.928             -3.111  1
                               (0.418)            (0.436)            (0.464)
Number of observations            2999               2999               5875  2
------------------------------------------------------------------------------
```

1. −2.93, −2.93, −3.11 with SE .418, .436, .464
2. design: all 5,875 rows with hours

QUICK CHECK

Why export with `etable ..., export()` rather than copy the table into Word?

A. it looks nicer

B. a rerun updates every number

C. Word cannot paste from Stata

ANSWER

Why export with `etable ..., export()` rather than copy the table into Word?

A it looks nicer

B a rerun updates every number

C Word cannot paste from Stata

The do-file is the source; the document is its output.

Part 5 on one page

CHEAT CARD · KEEP THIS ONE

`estimates store name`	keep a model
`estimates dir`	list stored models
`etable, estimates(a b)`	models side by side
`..., column(estimates) mstat(N)`	names and N
`..., export("f.docx", replace)`	to Word or Excel
`putdocx begin / text / image / save`	a note from the do-file
`graph export "f.png", replace`	a chart for the note

Practice on your own

PRACTICE SET · practice_D4.do, PART 5

CORE P1 Store the hours model with sex only as `h1` and with age group added as `h2`.

CORE P2 `etable` of `h1 h2`: type the coefficient for women in `h2`.

CORE P3 Export it to `output/p5_hours.xlsx`.

EXTENSION P4 A Word note with a title and one sentence.

CHALLENGE P5 Add the `marginsplot` of Part 4 to the note.

Each exercise has a CHECK line: silent means correct. Solutions are released at the end of the day.

Your model is now a table and a chart a manager can read.

lost? open `catchup_D4_S5.do`



6

DAY 4 · PART 6

The CPI, month by month

THE PROBLEM

Is inflation rising again?

WHAT WE HAVE

The consumer price index, all items and twelve divisions, every month since January 2007.

WHAT WE NEED

Inflation: the change over 12 months, and a chart of it.

WHY IT IS HARD

Each row needs the row 12 months earlier. Stata must know the rows are months.

Declare the series once; then one operator reaches back in time.

Part 6: a monthly series

AFTER THIS PART YOU CAN

- declare a monthly series with `tsset`
- chart it with `tsline`
- compute monthly and year-on-year inflation with `L.` and `L12.`
- check the latest figure against the NBS release

Colleagues in this room publish this series every month.

YOUR TURN · TASK 44

What is in the CPI files?

DO

Type:

```
describe using "data/cpi_divisions_long.dta"
```

YOU SHOULD SEE

Four variables: division, month, cpi, wbase. 3,068 rows.

IF NOT

The long CPI file

```
. describe using "data/cpi_divisions_long.dta"
    Variables:             4
division        byte    %35.0g     coicop     COICOP division (0 = all items)
month           int     %tm                   Month
cpi             double  %10.0g                Consumer price index (2014 = 100)
wbase           byte    %19.0g     wbase      Expenditure weights in force
------------------------------------------------------------------------------
Sorted by: division  month
```

1. 4 variables; 3,068 rows = 13 series × 236 months
2. 0 = all items, 1-12 = divisions
3. a monthly date, `%tm`
4. which survey's weights were in force

YOUR TURN · TASK 45

Open it, keep all items

DO

Type:

```
use "data/cpi_divisions_long.dta", clear
tabulate division if month == month[1]
keep if division == 0
list in 1/5
```

YOU SHOULD SEE

Thirteen series listed; 2,832 rows deleted; the first five months.

IF NOT

Thirteen series

```
. tabulate division if month == month[1]

    COICOP division (0 = all items) |      Freq.     Percent        Cum.
------------------------------------+-----------------------------------
                          All items |          1        7.69        7.69
  Food and non-alcoholic beverages |          1        7.69       15.38
    Alcoholic beverages and tobacco |          1        7.69       23.08
             Clothing and footwear |          1        7.69       30.77
Housing, water, electricity and gas |          1        7.69       38.46
  Furniture and household equipment |          1        7.69       46.15
                             Health |          1        7.69       53.85
                          Transport |          1        7.69       61.54
                      Communication |          1        7.69       69.23
              Recreation and culture |          1        7.69       76.92
                          Education |          1        7.69       84.62
              Restaurants and hotels |          1        7.69       92.31
   Miscellaneous goods and services |          1        7.69      100.00
------------------------------------+-----------------------------------
                              Total |         13      100.00
```

1. division 0: all items
2. twelve COICOP divisions

The all-items series

```
. keep if division == 0
(2,832 observations deleted)
. list in 1/5

     +-----------------------------------------------------+
     |  division     month         cpi                wbase |
     |-----------------------------------------------------|
  1. | All items   2007m1   46.413219   HBS 2006/07 weights |
  2. | All items   2007m2   46.420473   HBS 2006/07 weights |
  3. | All items   2007m3   46.795556   HBS 2006/07 weights |
  4. | All items   2007m4   46.848354   HBS 2006/07 weights |
  5. | All items   2007m5   46.896919   HBS 2006/07 weights |
     +-----------------------------------------------------+
```

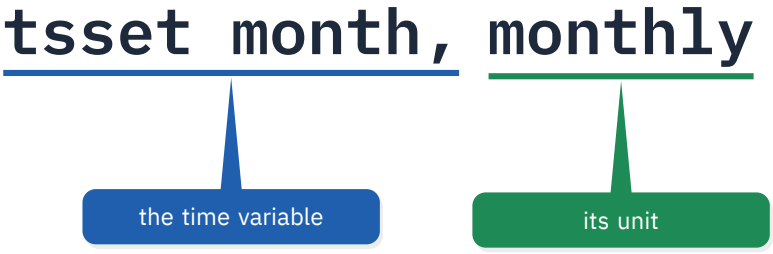
1. only all items left: 236 months
2. January 2007: 46.4 (2014 = 100)
3. the weights of the 2006/07 HBS

A monthly date Stata understands

what you see	what is stored
2007m1	564
2007m2	565
...	...
2026m8	799

months since January 1960. The format `%tm` shows them as dates; arithmetic works on the numbers.

Declare the series:



YOUR TURN · TASK 46

Declare it

DO

Type:

```
tsset month, monthly
```

YOU SHOULD SEE

The time variable and its range, one month apart.

IF NOT

Read the declaration

```
. tsset month, monthly

Time variable: month, 2007m1 to 2026m8 (1)
        Delta: 1 month (2)
```

1. January 2007 to August 2026
2. one month between rows

COMMON ERROR

Thirteen series, one time variable

```
. tsset month, monthly
repeated time values in sample
```

WHY

In the long file every month appears 13 times, once per division.

FIX

Keep one series first, or declare a panel (Part 7).

COMMON ERROR

An operator before tsset

```
. generate bad = L.cpi
time variable not set
```

WHY

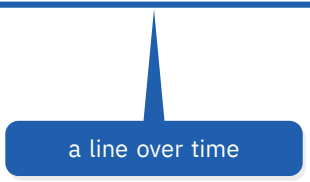
`L.` means ”the previous period”. Without `tsset` Stata does not know what a period is.

FIX

Run `tsset month, monthly` first.

Chart the series:

```
tsline cpi
```



YOUR TURN · TASK 47

Draw the index

DO

Type:

```
tsline cpi, title("Consumer price index, all items")
```

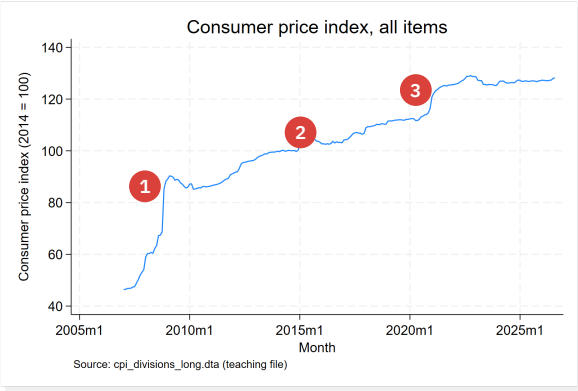
YOU SHOULD SEE

A rising line from 46 to 128.

IF NOT

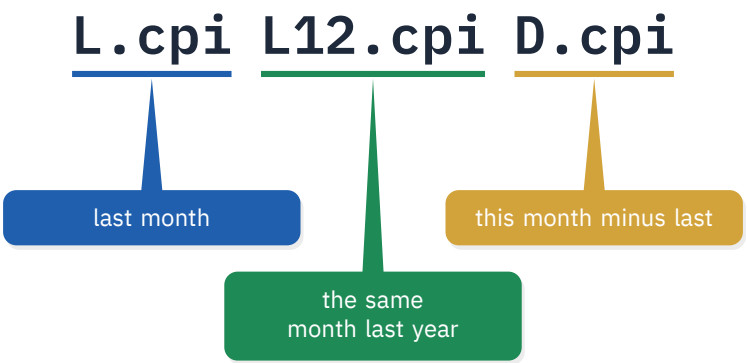
EXTRA Add a note with the source.

Read the index



1. end of 2008: prices jump
2. 2015: new weights (HBS 2013)
3. 2021: new weights (HBS 2018), and a sharp rise

Reach back in time:



Inflation in one line each

```
generate m_change = D.cpi
generate mom = 100 * (cpi / L.cpi - 1)
generate yoy = 100 * (cpi / L12.cpi - 1)
```

- points of index gained since last month
- month-on-month inflation, percent
- year-on-year inflation, percent: the figure NBS publishes

YOUR TURN · TASK 48

Compute inflation

DO

Type:

```
generate m_change = D.cpi
generate yoy = 100 * (cpi / L12.cpi - 1)
generate mom = 100 * (cpi / L.cpi - 1)
list month cpi m_change mom yoy in -13/l
```

YOU SHOULD SEE

1, 12 and 1 missing values generated; then the last 13 months.

IF NOT

EXTRA Why 12 missing for yoy?

The last 13 months

```
. list month cpi m_change mom yoy in -13/l
     +-------------------------------------------------------+
     |   month          cpi    m_change         mom         yoy |
     |-------------------------------------------------------|
224. |  2025m8   127.03623    .0412426    .0324757    .5470879 |
225. |  2025m9   126.96908   -.0671493   -.0528584    .5077931 |
226. | 2025m10   126.69924   -.2698443   -.2125276    .3048798 |
227. | 2025m11   126.99723    .2979874    .2351927    .0172556 |
228. | 2025m12   127.10614    .1089094    .0857573   -.2183335 |
     |-------------------------------------------------------|
229. |  2026m1    127.3128    .2066592    .1625879    .1006179 |
230. |  2026m2   127.21018   -.1026112   -.0805977     .296896 |
231. |  2026m3   127.16286   -.0473287   -.0372051    .2427256 |
232. |  2026m4   127.04953   -.1133258   -.0891186   -.0063574 |
233. |  2026m5   127.23998    .1904473    .1499001    .2861412 |
     |-------------------------------------------------------|
234. |  2026m6   127.32355    .0835688    .0656781    .4256209 |
235. |  2026m7   127.98448    .6609322    .5190966     .779155 |
236. |  2026m8   128.21595    .2314705    .1808582     .928645 |
     +-------------------------------------------------------+
```

1. August 2025: 127.04
2. July 2026: +.52% in one month
3. August 2026: 128.22; 0.93% over 12 months

NBS, August 2026 release: 128.22 and 0.93%.

The same figures as our two lines of Stata.

Check one published month every time you rebuild a series.

TRAP

Store differences of an index as `double`.

With `S12.cpi` stored as float, two months of 2008 disagree with `cpi - L12.cpi` beyond 1e-6. As double: none.

```
generate double s12 = S12.cpi
```

Float against double

```
. describe s12f s12d (1)

Variable      Storage   Display    Value
    name         type    format    label      Variable label
------------------------------------------------------------
s12f            float   %9.0g
s12d            double  %10.0g
. assert abs(s12f - (cpi - L12.cpi)) < 1e-6 if !missing(s12f)
2 contradictions in 224 observations
assertion is false
. assert abs(s12d - (cpi - L12.cpi)) < 1e-6 if !missing(s12d)
     +---------------------------------------------+
     |   month         cpi       s12f        s12d  |
     |---------------------------------------------|
 23. | 2008m11   85.207391    32.1414   32.141401 | (2)
 24. | 2008m12   88.213084   34.17802   34.178022 |
     +---------------------------------------------+
```

(1) float: 7 digits; double: 16

(2) 32.1414 against 32.141401: the float lost a digit

Chart inflation:

```
tsline yoy, yline(0)
```



YOUR TURN · TASK 49

The inflation chart

DO

Type:

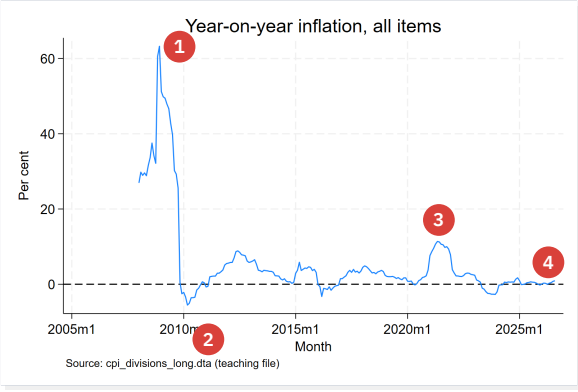
```
tsline yoy, yline(0) title("Year-on-year
inflation, all items") ///
    ytitle("Per cent") note("Source: NBS CPI")
```

YOU SHOULD SEE

A line from 2008 on, with a large peak and a dashed zero line.

IF NOT

Read the inflation chart



1. end of 2008: 63% in a year
2. 2010: prices falling
3. 2021: about 11%
4. August 2026: 0.9%

YOUR TURN · TASK 50

The latest month and the range

DO

Type:

```
summarize month
display %tm r(max)
list month yoy if month == r(max)
```

```
summarize yoy
```

YOU SHOULD SEE

The last month in words, its inflation, and the range of inflation.

IF NOT

Read the extremes

```
. display %tm r(max)
 2026m8
. list month yoy if month == r(max)

     +------------------+
     |  month       yoy |
     |------------------|
236. | 2026m8   .928645 |
     +------------------+
. summarize yoy

    Variable |        Obs        Mean    Std. dev.       Min        Max
-------------+---------------------------------------------------------
         yoy |        224    5.816064    11.76709  -5.538042   63.25156
```

1. the last month: August 2026
2. inflation from −5.5% to 63.3%

WARNING

Month-on-month changes carry seasonal patterns.

Fresh fish and fruit move with the season. Year-on-year compares like with like.

Publish year-on-year; read month-on-month with care.

A smoother trend:

```
tssmooth ma ma12 = yoy, window(11 1 0)
```

moving average into ma12

11 months back, this month, none ahead

YOUR TURN · TASK 51

Twelve-month average

DO

Type:

```
tssmooth ma ma12 = yoy, window(11 1 0)
list month yoy ma12 in -6/l
```

YOU SHOULD SEE

A description of the smoother, then six months.

IF NOT

Read the trend

```
. list month yoy ma12 in -6/l

     +-------------------------------+
     |  month         yoy       ma12 |
     |-------------------------------|
231. | 2026m3    .2427256    .282811 |
232. | 2026m4   -.0063574   .2757627 |
233. | 2026m5    .2861412   .2665033 |
234. | 2026m6    .4256209   .2619787 |
235. | 2026m7     .779155   .2736235 |
     |-------------------------------|
236. | 2026m8     .928645     .30542 |
     +-------------------------------+
```

1

1. inflation 0.93%; its 12-month average 0.31%

YOUR TURN · TASK 52

Any months missing?

DO

Type:

```
tsreport
```

YOU SHOULD SEE

Start, end, number of observations and gaps.

IF NOT

No gaps

```
. tsreport

Time variable: month
------------------------
Starting period = 2007m1
Ending period   = 2026m8
Number of obs   =    236
Number of gaps  =      0  (1)
```

(1) 236 months, no gap: every L12. has its month

ON YOUR OWN · J

Highest and lowest

THE QUESTION

What were the highest and the lowest year-on-year inflation rates since 2008?

HINT

```
summarize yoy.
```

ON YOUR OWN · J · ANSWER

```
summarize yoy
```

```
. summarize yoy

    Variable |        Obs        Mean    Std. dev.       Min        Max
-------------+---------------------------------------------------------
         yoy |        224    5.816064    11.76709   -5.538042   63.25156
```

1

1 63.3% and −5.5%

QUICK CHECK

Which line gives year-on-year inflation after `tsset month, monthly`?

A. generate y = 100*(cpi/L.cpi - 1)

B. generate y = 100*(cpi/L12.cpi - 1)

C. generate y = cpi - cpi[_n-12]

ANSWER

Which line gives year-on-year inflation after `tsset month, monthly`?

A. `generate y = 100*(cpi/L.cpi - 1)`

B. `generate y = 100*(cpi/L12.cpi - 1)`

C. `generate y = cpi - cpi[_n-12]`

A is month-on-month. C is a difference in index points, and breaks if a month is missing.

Part 6 on one page

CHEAT CARD · KEEP THIS ONE

`tsset month, monthly`	declare a monthly series
`tsline cpi`	chart it
`L.x L12.x D.x S12.x`	last month, last year, changes
`generate yoy = 100*(cpi/L12.cpi - 1)`	year-on-year inflation
`generate double s = S12.cpi`	keep the digits
`tssmooth ma m = x, window(11 1 0)`	12-month moving average
`tsreport`	gaps in the series

Practice on your own

PRACTICE SET · practice_D4.do, PART 6

CORE	P1	Year-on-year inflation for December 2025. Type it, two decimals.
CORE	P2	Month-on-month inflation for July 2026.
CORE	P3	Average year-on-year inflation in 2025.
EXTENSION	P4	How many months since 2008 had negative inflation?
CHALLENGE	P5	Inflation over 24 months to August 2026 (`L24.`).

Each exercise has a CHECK line: silent means correct. Solutions are released at the end of the day.

You can declare a time series, compute inflation and chart it.

lost? open **catchup_D4_S6.do**



7

DAY 4 · PART 7

Many series at once

THE PROBLEM

Which prices are driving inflation?

WHAT WE HAVE

Thirteen series in one long file: one row per division and month.

WHAT WE NEED

Inflation inside each division, without one series leaking into the next.

WHY NOT TSSET

Each month appears 13 times (the error in Part 6).

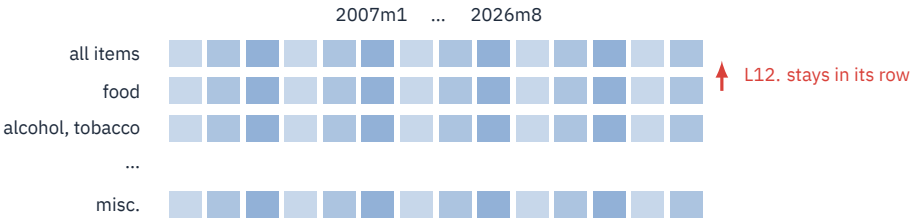
Declare a panel: a series per division.

Part 7: a panel of series

AFTER THIS PART YOU CAN

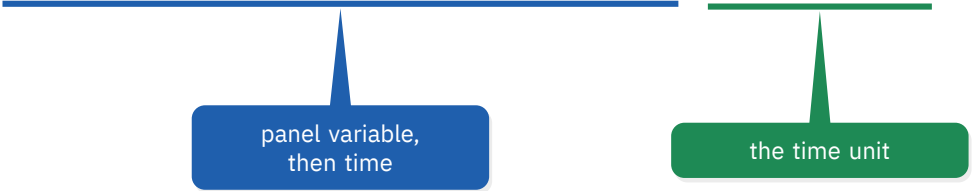
- declare a panel with `xtset`
- compute inflation inside each division
- chart several divisions with `xtline`
- find the division with the highest inflation

One series per division



Declare the panel:

```
xtset division month, monthly
```



YOUR TURN · TASK 53

Declare it

DO

Type:

```
use "data/cpi_divisions_long.dta", clear
xtset division month, monthly
```

YOU SHOULD SEE

The panel variable, the time variable, ”strongly balanced”.

IF NOT

Read the declaration

```
. xtset division month, monthly

Panel variable: division (strongly balanced) 1
 Time variable: month, 2007m1 to 2026m8 2
         Delta: 1 month
```

1. every division has every month
2. 2007m1 to 2026m8

YOUR TURN · TASK 54

Inflation inside each division

DO

Type:

```
generate yoy = 100 * (cpi / L12.cpi - 1)
list division month cpi yoy ///
    if month == tm(2026m1) & division <= 4
```

YOU SHOULD SEE

156 missing values (12 per series); then five rows for January 2026.

IF NOT

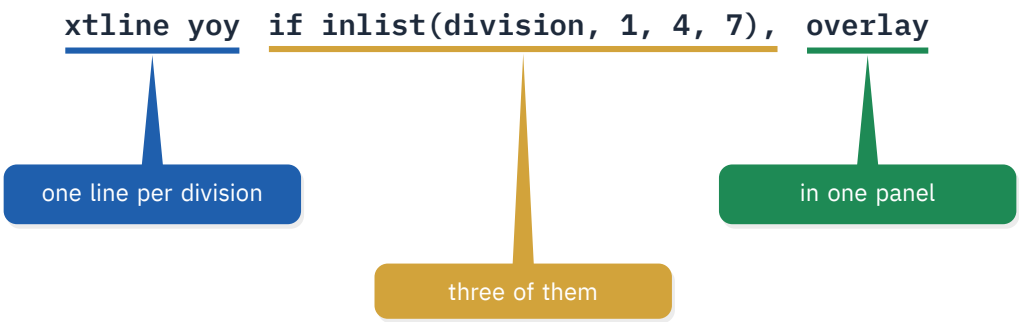
EXTRA Why 156?

January 2026, five series

```
. generate yoy = 100 * (cpi / L12.cpi - 1)
(156 missing values generated)
      +-------------------------------------------------------------------+
      |                          division    month         cpi         yoy |
      |-------------------------------------------------------------------|
 229. |                         All items   2026m1    127.3128    .1006179 |
 465. |    Food and non-alcoholic beverages   2026m1   121.30794    .2548792 |
 701. |     Alcoholic beverages and tobacco   2026m1   153.47325    2.865528 |
 937. |                Clothing and footwear   2026m1   104.55879   -.4310083 |
1173. | Housing, water, electricity and gas   2026m1   140.87062   -1.669359 |
      +-------------------------------------------------------------------+
```

1. 12 per series: L12. never reaches into another division
2. alcohol and tobacco: +2.9%
3. housing and utilities: −1.7%

Several series in one chart:



YOUR TURN · TASK 55

Food, housing, transport

DO

Type:

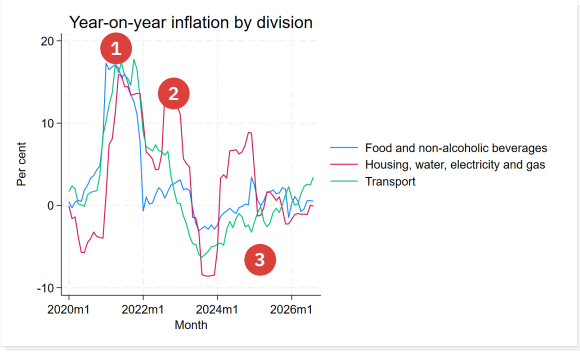
```
xtline yoy if inlist(division, 1, 4, 7) ///
    & month >= tm(2020m1), overlay ///
    title("Year-on-year inflation by division")
ytitle("Per cent")
```

YOU SHOULD SEE

Three lines from 2020, one colour each, with a legend.

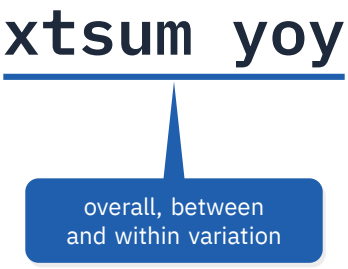
IF NOT

Read the three divisions



1. 2021: food and transport near 17%
2. housing peaks again in 2022
3. 2023-2024: all three negative

Across divisions and over time:



YOUR TURN · TASK 56

Split the variation

DO

Type:

```
xtsum yoy
```

YOU SHOULD SEE

Three rows: overall, between, within.

IF NOT

Read the split

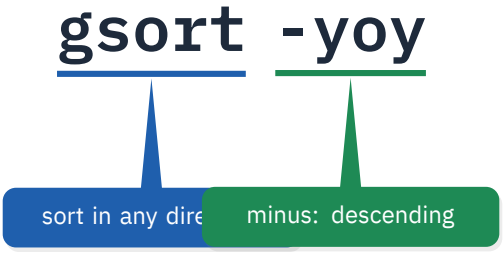
```
. xtsum yoy

Variable          |      Mean   Std. dev.       Min        Max |    Observations
------------------+--------------------------------------------+----------------
yoy       overall |  5.384555   11.63305  -20.52511   89.53938 |     N =    2912
          between |             1.610071   1.631771   7.383834 |     n =      13  (1)
          within  |              11.5297  -22.52439    87.5401 |     T =     224  (2)
```

(1) between divisions: SD 1.6, long-run averages are close

(2) within a division over time: SD 11.5, most of the movement

Sort, largest first:



ON YOUR OWN · K

Which prices drive inflation now?

THE QUESTION

In August 2026, which three divisions had the highest year-on-year inflation?

HINT

`keep if month == r(max)`, then `gsort -yoy`.

ON YOUR OWN · K · ANSWER

```
summarize month

keep if month == r(max) & division > 0

gsort -yoy

list division yoy in 1/3
```

```
     +-----------------------------------------+
     |                        division       yoy |
     |-----------------------------------------|
  1. |                       Transport  3.456891 |
  2. |         Restaurants and hotels  3.427587 |
  3. | Alcoholic beverages and tobacco  2.911193 |
     +-----------------------------------------+
```

1

1 transport 3.5%, restaurants 3.4%, alcohol and tobacco 2.9%

Panel models (fixed and random effects) are the next step after this course.

QUICK CHECK

After `xtset division month`, what is L12.cpi in the first row of Food?

A. all items 12 months earlier

B. missing: there is no earlier Food month

C. the last row of division 0

ANSWER

After `xtset division month`, what is L12.cpi in the first row of Food?

A all items 12 months earlier

B missing: there is no earlier Food month

C the last row of division 0

Operators stay inside their panel: 12 missing per series (log X02).

Part 7 on one page

CHEAT CARD · KEEP THIS ONE

`xtset division month, monthly`	declare a panel
`L12.cpi`	last year, same division
`if month == tm(2026m1)`	a monthly date in a condition
`xtline y, overlay`	one line per series
`xtsum y`	between and within
`gsort -y`	largest first

Practice on your own

PRACTICE SET · practice_D4.do, PART 7

CORE	P1	Year-on-year inflation of Education in August 2026.
CORE	P2	Which division had the lowest inflation in August 2026?
CORE	P3	Health: highest year-on-year inflation since 2008. Type it.
EXTENSION	P4	How many divisions had negative inflation in August 2026?
CHALLENGE	P5	Chart transport and all items together since 2020.

Each exercise has a CHECK line: silent means correct. Solutions are released at the end of the day.

You can handle a set of NBS series as one panel.

lost? open `catchup_D4_S7.do`

Lab 2 · The analytical note

THE SITUATION

The Minister's office wants two pages by tomorrow: who is more likely to be unemployed, and where inflation stands.

OPEN

`lab_D4_2.do` in `C:\NBS_Stata\day4`. Close Word before each run.

THE TASKS

CORE Tasks 1 to 5: design-based model, predicted rates, chart, inflation, the note

EXTENSION Task 6: add the model table

CHALLENGE Task 7: a sentence naming the top division, from the data

CORRECT WHEN

the file runs without red text and `output/note_q1.docx` opens.

The lab file: lab_D4_2.do

```
* TASK 3 (Core)  Predicted rates and the chart
*   Predicted unemployment rate by age group; copy r(table) into M at
*   once. Chart it with marginsplot (title, source note) and export it
*   to output/D4_margins.png.
* ------------------------------------------------------------------------
* >>> YOUR CODE



* CHECK Task 3 (silent = correct)
```

Close Word before each run: Task 5 writes `output/note_q1.docx`.

Green note when your last check is silent. Then the Challenge task.

Day 4, done

YOU CAN NOW

- test a difference the way NBS should report it
- fit and read regressions with categories
- check a model and pick its standard errors
- give predicted rates and differences in points
- put models in a table and a note
- compute and chart inflation, series by series

TODAY'S PRODUCT

`note_q1.docx`

TOMORROW

Everything from this week runs from one master file.

Daily check

1. Open `check_D4.do` from the day4 folder.
2. Type your **participant code** on the first line.
3. One sentence: what was unclear today?
4. Close Word. Save and run with **Ctrl + D**.

Read the last line: how many of the checks are done. It is not a grade.

Tomorrow everything you built this week runs from one file.